

CEO Turnovers and Disruptions in Customer–Supplier Relationships

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Abstract

Events that disrupt customer–supplier relationships pose a source of risk for suppliers that depend on a customer for a large portion of their revenues. We identify the replacement of a customer’s chief executive officer (CEO) as a disruptive event that results in suppliers losing substantial sales. These losses are greater when an incumbent customer CEO is more likely to be entrenched and stem largely from the successor divesting assets. Finally, we document that losses in sales following a customer CEO turnover lead to declines in a supplier’s financial performance and that suppliers experience negative abnormal stock returns to announcements of customer CEO departures.

I. Introduction

Prior literature suggests that suppliers deriving much of their revenues from a few customers are more profitable due to greater operational efficiencies (Kalwani and Narayandas (1995), Patatoukas (2012), and Cen, Dasgupta, and Sen (2016)). However, these dependent suppliers are less diversified and often must invest in assets that offer little value outside of these relationships. Consequently, disruptions in the relationships represent a significant source of risk for dependent suppliers and their investors. While regulators require dependent suppliers to

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acknowledge this risk, there is surprisingly little direct evidence identifying events that can disrupt these relationships.¹

In this study, we investigate whether the replacement of a customer firm's chief executive officer (CEO) affects its relationships with dependent suppliers.² A large stream of research documents that CEO turnovers can often prompt strategic and operational changes within firms. New CEOs often divest assets and divisions (Weisbach (1995), Berry, Bizjak, Lemmon, and Naveen (2006), and Pan, Wang, and Weisbach (2016)), cut capital expenditures and employment (Denis and Denis (1995)), reallocate resources across business segments (Berger and Ofek (1999), Xuan (2009)), and adjust corporate policies (Huson, Malatesta, and Parrino (2004), Cronqvist, Makhija, and Yonker (2012), and Dittmar and Duchin (2016)). Given the dramatic firm-level changes that can occur, customer CEO turnovers could have a significant impact on suppliers. The direction of this effect, however, is unclear. On the one hand, if new CEOs divest divisions that suppliers sell to or reduce the scope of their firms' operations, suppliers that rely on the customer for a large portion of revenues will likely suffer losses in sales. Suppliers could also lose substantial business if a successor switches suppliers, as was the case at Apple Inc. when Steve Jobs' successor, Tim Cook, shifted some production from the company's long-time supplier Foxconn to Pegatron in part to lower costs, expand product lines, and diversify risk (Dou (2013)).

On the other hand, the replacement of a customer's CEO may benefit dependent suppliers. If new CEOs restructure their firms, suppliers could profit from increased sales if they sell to a division that receives an inflow of resources. Successors also are more likely to try to renegotiate contracts and demand better terms of sale. This can indirectly help some suppliers if they gain new business at the expense of other suppliers that are reluctant to accede to a new CEO's demands. Finally, if CEO turnovers generate improvements in operating performance, suppliers should benefit from working with a customer with more sustainable profits and a higher likelihood of future growth.

To examine whether and how the replacement of a customer's CEO affects its relationships with dependent suppliers, we identify public suppliers that depend on business from an ExecuComp-listed firm over the 1993 to 2011 period. We define dependent suppliers as those that rely on a customer for a large portion of their revenues or report a customer as important to their business.³ From this sample, we identify dependent suppliers with customers that replace their CEOs

¹A few studies provide indirect evidence of specific events that negatively affect suppliers. For instance, suppliers experience negative abnormal returns when customers declare bankruptcy (Hertzel, Li, Officer, and Rodgers (2008), Kolay, Lemmon, and Tashjian (2016)), engage in horizontal mergers (Fee and Thomas (2004), Shahrur (2005)), and participate in leveraged buyouts (Brown, Fee, and Thomas (2009)).

²Throughout the paper, we use the terms "customer firm" and "customer" interchangeably and in reference to the customer firm of existing dependent suppliers. We also use the terms "dependent suppliers" and "suppliers" interchangeably and in reference to suppliers that depend on one or more important customers.

³Regulations require suppliers to disclose customers that individually account for 10% or more of the supplier's revenues. However, some suppliers voluntarily identify customers accounting for less than 10% of sales if the customer is important to their business (Patatoukas (2012)). Thus, we define any supplier that explicitly discloses the name and sales of an individual customer as a dependent supplier.

and examine whether suppliers' sales to customers change following CEO departures. If customer CEO turnovers hurt (help) relationships with suppliers, we should observe a decline (increase) in supplier sales to customers that replace their CEOs.

We adopt a difference-in-differences research design and use panel regressions, controlling for customer–supplier fixed effects, year fixed effects, and several supplier and customer characteristics. Overall, we find evidence consistent with CEO turnovers disrupting relationships with suppliers. Following the replacement of a customer's CEO, the average supplier loses \$19.1 million in sales to that customer and the likelihood of the business relationship ending increases by 12%. The average supplier also sees a total sales drop of 2.6% following the customer CEO departure. Furthermore, suppliers experience significantly negative cumulative abnormal stock returns (CARs) when customers announce that they are replacing their CEOs.

Two key characteristics of customer–supplier relationships that could affect the magnitude of supplier losses following a customer CEO turnover are i) the customer's cost of switching suppliers and ii) whether a supplier makes relationship-specific investments (RSIs). If customer CEO turnovers disrupt relationships with suppliers, we expect a smaller drop in sales to customers that face higher switching costs. Using the fraction of a customer's total inputs provided by a supplier, the number of years that a customer has been purchasing from a supplier, and a supplier's market share and size as proxies for switching costs (Porter (1980), Schumacher (1991), Snyder (1996), Inderst and Wey (2007), Brown et al. (2009), and Hui, Klasa, and Yeung (2012)), we find that higher switching costs mitigate the decline in supplier sales following customer CEO turnovers.

The effect of RSIs on the negative relation between customer CEO turnovers and supplier sales is less straightforward. RSIs typically involve unique or specialized products that offer little value outside of these relationships and can arise from suppliers producing specialized inputs for research and development (R&D) intensive customers (Levy (1985), Titman and Wessels (1988), and Raman and Shahrur (2008)). Thus, RSIs can increase a customer's cost of switching suppliers. However, a supplier that has made RSIs may not have the production capabilities and flexibility to begin producing goods that fit the customer's new demands. We use supplier and customer R&D intensity to proxy for the likelihood that suppliers have made RSIs (Allen and Phillips (2000), Kale and Shahrur (2007)). We find larger losses in sales when suppliers have likely made RSIs, suggesting that CEO departures lead to greater disruptions when suppliers have limited production flexibility.

While our results suggest that successors make significant changes that disrupt relationships with dependent suppliers, the motivation for these changes can arise from several channels. New CEOs may adjust corporate policies and restructure assets to more closely reflect their preferences and skills, which may not be the same as those of incumbents (Cronqvist et al. (2012), Dittmar and Duchin (2016)). Alternatively, changes could arise, in part, from firms viewing the replacement of their CEOs as necessary when pursuing new strategies, as successors do not have vested interests in prior strategies and are more willing to recognize the need for improvement (Sliwka (2007), Casamatta and Guembel (2010)).

While the former explanation may be unrelated to the incumbent's performance, the latter explanation suggests that the incumbent may be entrenched and that current policies are suboptimal.

To shed light on these channels, we conduct two sets of cross-sectional tests, beginning with an examination of whether the nature of the turnover affects the negative relation between the CEO departure and supplier sales. Prior work suggests that boards often hire outsiders when they want to break with the failed policies of incumbents (Borokhovich, Parrino, and Trapani (1996), Parrino (1997)). Similarly, forced departures imply that incumbents were not acting in the best interests of shareholders or that their strategies failed. Therefore, larger declines in supplier sales following outsider successions or forced turnovers would suggest that our findings stem, in part, from successors amending or severing an incumbent's suboptimal strategies. Consistent with this notion, we document that outsider appointments and forced departures lead to larger losses in supplier sales. We also find that suppliers experience more negative CARs to announcements of customer CEO turnovers when outsiders are appointed or incumbents are forced out.

Outside successions and forced turnovers, however, do not always indicate that incumbents were entrenched. For instance, outsider appointments could be associated with the board wanting to implement strategic changes in areas beyond an incumbent's expertise, and forced turnovers may simply result from a poor fit between an incumbent's skills and those required for the job. Thus, in a second analysis, we more directly test whether predecessor entrenchment helps explain our results by examining whether the decline in supplier sales is more pronounced when the incumbent has relatively high discretionary decision-making power over the board or when the board is less likely to monitor the CEO. We find that declines in sales are larger when the incumbent has more power and when the firm has low director share ownership or a busy board of directors. Suppliers also experience more negative CARs to the announcement of a customer CEO turnover when the incumbent has more power. Overall, these findings suggest that one reason turnovers disrupt relationships with suppliers is that successors change failing strategies, which would also explain why changes arise only after a new CEO takes office.

Next, we investigate the mechanisms that result in lower supplier sales and how these affect the supplier's overall financial performance. Declines in supplier sales following customer CEO turnovers could stem from factors such as successors divesting assets, reallocating resources, switching suppliers, or demanding better terms of sales. Our results suggest that, when suppliers lose sales following CEO turnovers, successors in their first 3 years in office are more likely to divest assets, lower total purchases from all dependent suppliers, and have fewer suppliers disclose their firms as important customers. However, we do not observe a change in customers' gross margins. These findings suggest that much of the decline in sales can be explained by successors divesting assets rather than reallocating resources to other segments, switching suppliers, or receiving better pricing. Finally, we document that losses in supplier sales to customers following CEO departures are associated with declines in supplier firm value, return on assets, profit, operating, and gross margins, and increases in selling expenses.

While we gain valuable insights from examining how all types of customer CEO turnovers affect customer–supplier relationships, several turnovers are endogenously determined. For instance, in the case of forced turnovers, the board of directors recognizes the need for change and hires a new CEO. Thus, it is not the turnover per se that causes disruptions. Rather, the cause is the need for change, and turnover is how the board implements that change. In our last set of analyses, we address endogeneity concerns. We show that the negative relation between CEO turnovers and supplier sales is robust to using a propensity score matched sample. Our findings are also robust to using a subsample of exogenous CEO departures, defined as those resulting from the death or illness of outgoing CEOs or natural retirements (Fee, Hadlock, and Pierce (2013)). For this set of exogenous turnovers, we also show that sales to customers decline only after the turnover and not before. Collectively, these results suggest that, even following plausibly exogenous turnovers, successors make strategic and operational changes that disrupt their relationships with suppliers.

The central contribution of our study is that we provide both indirect and direct evidence that customer CEO turnovers disrupt relationships with dependent suppliers. Our indirect evidence comes from documenting negative market reactions for these suppliers around the announcement of a customer CEO turnover. Prior work also provides indirect evidence of other disruptive events by showing that suppliers experience negative CARs when customers declare bankruptcy (Hertzel et al. (2008), Kolay et al. (2016)), engage in horizontal mergers (Fee and Thomas (2004), Shahrur (2005)), and participate in leveraged buyouts (Brown et al. (2009)). Our announcement return evidence suggests that the level of disruption resulting from a customer CEO departure is similar in magnitude to that of an arguably more obvious disruption: the bankruptcy of a major customer. We observe average supplier CARs ranging from -0.61% to -2.73% when customers announce the replacement of their CEOs, which is in line with observed CARs of -1.94% when customers file for bankruptcy (Hertzel et al. (2008)). Importantly, our study differs from most previous work that focuses on market reactions in that we also provide direct evidence that CEO turnovers are disruptive from an examination of supplier sales following customer CEO departures. Specifically, we show that suppliers experience declines in sales to customers that replace their CEOs and that the loss in sales is associated with lower overall financial performance. In sum, we conclude that customer CEO turnovers represent a significant source of risk for dependent suppliers and their investors.

Our findings also contribute to the stream of literature that examines how customer–supplier relationships impact suppliers. These relationships affect financial leverage choices (Kale and Shahrur (2007), Banerjee, Dasgupta, and Kim (2008)), trade credit policies (Banerjee, Dasgupta, and Kim (2007), Dass, Kale, and Nanda (2015)), cash holdings (Itzkowitz (2013)), payout policies (Wang (2012)), financial constraints and investment (Itzkowitz (2015)), accounting choices (Raman and Shahrur (2008), Hui et al. (2012)), and governance practices (Johnson, Karpoff, and Yi (2015)). They also affect risk and financing costs (Albuquerque, Papadakis, and Wysocki (2014), Cen, Dasgupta, Elkamhi, and Pungaliya (2016), Dhaliwal, Judd, Serfling, and Shaikh (2016), and Campello and Gao (2017)) as well as performance and valuation (Kalwani and Narayandas (1995),

Johnson, Kang, and Yi (2010), Bhattacharyya and Nain (2011), Patatoukas (2012), Ak and Patatoukas (2016), Cen, Dasgupta, and Sen (2016), and Irvine, Park, and Yildizhan (2016)). Thus, our finding that the replacement of a customer's CEO disrupts relationships with suppliers implies that customer CEO turnovers can also impact a supplier's corporate policies.

Last, our study extends work examining the outcomes of CEO turnovers. Most prior research shows that CEO turnovers lead to increases in shareholder wealth and improvements in operating performance (Denis and Denis (1995), Huson et al. (2004), and Berry et al. (2006)). There is, however, little research investigating whether CEO turnovers impact nonfinancial stakeholders. We provide evidence that CEO turnovers negatively affect one such nonfinancial stakeholder: dependent suppliers.

The remainder of this paper is organized as follows: Section II develops our principal hypothesis and predictions. Section III describes the sample construction, empirical methodology, and data. Section IV presents our empirical findings. Section V addresses robustness tests. Section VI concludes.

II. Hypothesis Development

Prior research identifies CEO turnovers as events that initiate and signal substantial strategic and operational changes within firms. For instance, Pan et al. (2016) find that successors are more likely to divest and discontinue operations in their first few years in office, and Weisbach (1995) documents that new CEOs are more likely to divest poorly performing acquisitions made by incumbents. Several other studies also associate corporate downsizing with CEO turnovers by showing that new CEOs are more likely to divest assets, reduce capital expenditures, and cut employment (Denis and Denis (1995), Berger and Ofek (1999), Huson et al. (2004), and Berry et al. (2006)). In addition, CEO turnovers can lead to changes in corporate policies due to differences in personal traits and preferences between successors and incumbents. For example, Cronqvist et al. (2012) show that successors with higher personal leverage preferences increase corporate debt ratios, and Dittmar and Duchin (2016) find that more cautious CEOs increase their firms' cash holdings. Xuan (2009) also documents that incoming CEOs often reallocate resources across business segments to gain the favor of certain divisions.

Applying these findings to our study, strategic and operational shifts following the replacement of a customer firm's CEO will likely hurt some dependent suppliers but benefit others. For example, suppliers could lose sales if successors divest divisions that a supplier sells to or reduce the scope of their firms' operations. However, suppliers of divisions that new CEOs do not downsize may benefit from a transfer of resources from downsized divisions. Also, a supplier will lose substantial business if a successor switches to a different supplier, whereas the supplier that receives the new business will see a considerable increase in revenues.

Anecdotal evidence further suggests that new CEOs are more likely to try to renegotiate terms of sales and, in particular, prices. For example, shortly after his appointment as CEO of Chrysler Group, Dieter Zetsche set a strategic goal of cutting supplier costs by at least 15% over the following 3 years (Ball (2001)).

Nissan's new CEO, Carlos Ghosn, also reduced costs dramatically by severing relationships with suppliers and forcing them to rebid on price (Pilling (2007)). Thus, another way CEO turnovers can impact a customer's relationships with its suppliers is through renegotiations or contract and sales-term amendments. Suppliers will likely suffer if new CEOs can secure lower prices, or suppliers may lose business altogether if they will not accept more onerous supply terms. At the same time, other suppliers will benefit if they gain new contracts at the expense of suppliers that are reluctant to accept the demands of new CEOs.

Last, suppliers that rely on an important customer depend on the prospects of such business relationships and are, thus, exposed to changes in the customer's financial condition (Titman and Wessels (1988), Banerjee et al. (2008)). In particular, financially distressed customers are more likely to be liquidated, which can hurt supplier sales to these customers. On the other hand, prior studies often associate CEO turnovers with improvements in operating performance (Denis and Denis (1995), Huson et al. (2004)), which can boost supplier sales. In this scenario, all suppliers could benefit from a customer with more sustainable profits and a higher likelihood of growth.

In sum, both empirical and anecdotal evidence suggests that the replacement of a customer's CEO can have a significant impact on sales from its dependent suppliers. Our discussion highlights two competing effects of customer CEO turnovers on sales from suppliers, as some suppliers may lose while others gain following the turnover event. Thus, whether CEO turnovers impact sales from these suppliers, on average, is ultimately an empirical question, which leads to our principal hypothesis:

Hypothesis. The replacement of a customer's CEO disrupts its relationships with dependent suppliers.

We measure disruptions in customer-supplier relationships by examining changes in sales to a customer from a dependent supplier. If CEO turnovers disrupt these relationships and hurt suppliers on average, our hypothesis generates the empirical prediction that supplier sales to customers that replace their CEOs should decline. However, if more suppliers benefit than are hurt by CEO departures, we would expect sales to increase following turnovers. To the extent that these two effects work against each other, we may find no significant relation between customer CEO turnovers and sales from suppliers.

III. Data and Empirical Methodology

A. Identifying Supplier Sales to Customers

We use Compustat's segment customer files to identify suppliers that are incorporated in the United States and that report customers during the 1993–2011 period. Since 1976, the Statement of Financial Accounting Standards No. 14 (FAS 14) of the Financial Accounting Standard Board (FASB) has required suppliers to disclose external customers that individually account for 10% or more of the supplier's revenues. FAS 131 superseded FAS 14 in 1997, but the requirement to report such customers remains for public companies under The U.S. Securities and Exchange Commission (SEC) Regulation S-K Item 101. Suppliers also

often voluntarily report customers that account for less than 10% of sales, and we include these observations in our analysis. We exclude suppliers in utilities (Standard Industrial Classification (SIC) codes 4900–4999) and financial industries (SIC codes 6000–6999). Because customers do not have a unique identifier linking them to Compustat, we next match reported customer names to firms listed on ExecuComp to obtain data on customer-firm CEOs. We are able to match approximately 47.1% of reported customers to the ExecuComp database. We exclude customers that are utilities, financial firms, or foreign firms and instances when the supplier or customer is a subsidiary of the other.

To map supplier sales to specific customers, we match each ExecuComp customer to its suppliers from the first year a supplier reports sales to the customer to 2 years after the customer–supplier relationship ends.⁴ We extend the panel to 2 years after the last year the supplier reports sales to the customer and set sales in these years to 0, which helps account for the relationship deteriorating rather than just disappearing.⁵ For years when sales to customers are missing between the first and last years of relationships, we set sales to 0.

B. Identifying CEO Turnovers and Creating the Final Sample

Using ExecuComp, we next identify 597 instances when a customer with a dependent supplier replaces its CEO. If a customer replaces its CEO more than once during a particular customer–supplier relationship, we drop all customer–supplier observations after the first successor’s tenure. We set the indicator variable POST_TURNOVER to 1 for the years following the turnover event. Specifically, POST_TURNOVER takes a value of 1 for all years during a successor’s tenure, and 0 otherwise. Thus, our panel data set includes a clearly defined pre and post period so that we can attribute a decline in supplier sales to the replacement of a particular CEO. For customer–supplier relationships that experience a turnover event, the median number of years in both the pre and post period is 4. Extending our analysis beyond the year of a turnover allows time for purchase obligations to expire (Williams (2012)) and new CEOs to change strategies, divest assets, switch suppliers, and/or renegotiate agreements.

Last, we obtain financial statement data from the Compustat annual files, stock return data from the Center for Research in Security Prices (CRSP) files, and blockholder data from Thomson Reuters Insider Filing Data Feed. After merging the databases and removing observations with missing values for our variables, the main sample consists of 22,797 customer–supplier years, 5,876 unique

⁴The starting year of Compustat’s segment database is 1976. Thus, the first year that we can identify a relationship is 1976. Nevertheless, our results should not be biased by this truncation as only 0.08% of relationships in our sample started in 1976. In addition, because 1993 is the first year that we have ExecuComp data available, our panel construction is less sensitive to this truncation.

⁵We select 2 years because the median customer–supplier relationship in our sample lasts 2 years. In Section IV.A, we show that our results are robust to matching each customer to its suppliers: i) from the first year of the customer–supplier relationship to 1 year after the relationship ends and setting sales to 0 in this last year, ii) beginning 2 years before the first year of the customer–supplier relationship to 2 years after the relationship ends and setting sales to 0 in the 2 years before and after the relationship, and iii) beginning 1 year before the first year of the customer–supplier relationship to 1 year after the relationship ends and setting sales to 0 in the 1 year before and after the relationship. These latter two tests help account for the relationships ramping up rather than just appearing.

customer–supplier pairs, 2,998 unique suppliers, and 787 unique customers. Of the 5,876 customer–supplier pairs, 2,586 (44.0%) have a customer that replaces its CEO. These form our treatment group. The remaining 3,290 (56.0%) customer–supplier pairs have a customer that never replaces its CEO. These are our control group.

Table 1 reports the top 20 customers with the most dependent suppliers. In general, large firms selling to end consumers, such as Walmart, General Motors, AT&T, Ford, and Chrysler, comprise this list. In our sample, the mean (median) customer firm has 2.75 (1.00) dependent suppliers in any given year. We examine the robustness of our findings to controlling for customers that have many dependent suppliers in Table 4.

C. General Empirical Methodology

We adopt a difference-in-differences research design to examine the relation between customer CEO turnovers and supplier sales to these customers. Customers replace their CEOs at different times during the sample period. Thus, we examine the before–after effect in customer–supplier relationships in which the customer replaces its CEO (the treatment group) vis-à-vis the before–after effect in relationships in which the customer does not replace its CEO (the control group). We implement this test through the following panel regression:

$$(1) \ln(1 + \text{PCT_SALES})_{ict} = \alpha_1(\text{POST_TURNOVER})_{ict} + X_{ict}\beta + v_t + \omega_j + \varepsilon_{ict},$$

where PCT_SALES_{ict} is sales from supplier i to customer c in year t scaled by the supplier's total sales in year t (in percent). POST_TURNOVER is an indicator variable set to 1 if a customer has replaced its CEO during the customer–supplier relationship as of year t , and 0 otherwise. X_{ict} is a set of control variables, v_t

TABLE 1
Customers with the Most Dependent Suppliers

Table 1 reports the names and industries of the 20 public customers that have the most dependent suppliers in a given year using public suppliers and customers identified from Compustat and ExecuComp over 1993–2011. For each customer, the number of dependent suppliers is tabulated each year and then averaged over the entire sample period for which the customer has data available.

Customer Name	2-Digit SIC Industry	Average No. of Dependent Suppliers
Walmart Stores	53 - General Merchandise Stores	91.8
General Motors	37 - Transportation Equipment	57.3
International Business Machines	73 - Business Services	43.3
AT&T	48 - Communication	40.8
Chrysler	37 - Transportation Equipment	36.5
Ford Motor	37 - Transportation Equipment	30.5
Motorola	36 - Electronic and Other Electric Equipment	25.6
AmerisourceBergen	51 - Wholesale Trade (Nondurable Goods)	22.4
Cardinal Health	51 - Wholesale Trade (Nondurable Goods)	21.8
Hewlett Packard	35 - Industrial Machinery and Equipment	20.6
Compaq Computer	35 - Industrial Machinery and Equipment	20.0
General Electric	36 - Electronic and Other Electric Equipment	19.9
Verizon Communications	48 - Communication	19.4
McKesson	51 - Wholesale Trade (Nondurable Goods)	18.9
Lucent Technologies	73 - Business Services	17.7
Sears Holdings	53 - General Merchandise Stores	17.4
Target	53 - General Merchandise Stores	16.8
Sprint Nextel	48 - Communication	16.3
Intel	48 - Communication	16.2
Digital Equipment	35 - Industrial Machinery and Equipment	15.5

are year fixed effects, and ω_j are customer–supplier fixed effects. We correct for heteroskedasticity and serial correlation within supplier groupings by clustering standard errors at the supplier level.⁶ The year fixed effects account for transitory nationwide factors, such as macroeconomic conditions, that may affect supplier sales to customers and the likelihood that a customer firm replaces its CEO. The customer–supplier fixed effects control for time-invariant omitted customer–supplier characteristics that could differ across customer–supplier relationships (e.g., if explicit contracts govern certain relationships, while implicit agreements form the basis of other relationships). The customer–supplier fixed effects also ensure that estimates for α_1 reflect actual changes in sales to a customer from a specific dependent supplier over time rather than simple cross-sectional correlations. Thus, α_1 is the difference-in-differences estimator that measures the *within* customer–supplier pair sales differences *before* and *after* the CEO turnover event vis-à-vis similar before-after sales differences in customer–supplier pairs that do not experience a CEO turnover during the same period.

It is helpful to consider an example. Suppose we want to estimate the effect of Walmart (the customer) replacing its CEO in the year 2000 on sales from Clorox (the supplier). We can subtract the level of sales from Clorox to Walmart before the turnover from the level of sales after the turnover. However, economy- or industry-wide shocks may occur simultaneously and affect sales in the year 2000. Furthermore, if there is a natural decay rate in sales from a dependent supplier to a customer over time, then the difference between pre- and post-turnover sales may be biased downward. To control for such factors, we calculate the same difference in a control customer–supplier relationship, say between Walgreens (the customer) and Cardinal Health (the supplier), in which the customer does not replace its CEO during the year 2000. Finally, we can calculate the difference of these two differences, which represents the incremental effect of the CEO turnover on the customer–supplier relationship in the treatment group (Walmart and Clorox) compared to the effect on the relationship in the control group (Walgreens and Cardinal Health).

The tests used in this paper are more stringent than the simple intuition provided above because we control not only for customer–supplier relationship differences but also for other customer and supplier-specific unobservable and observable differences. Another advantage is that customers replace their CEOs at different times, which allows customer–supplier pairs to be in both our treatment group (if the customer has already replaced its CEO) and control group (if the customer has not yet replaced its CEO).

D. Further Discussion of Methodology to Measure Supplier Sales to Customers

Although suppliers often report customers that account for less than 10% of their total sales, accounting and security regulations require suppliers to report only customers that account for at least 10% of total sales. Consequently, when a supplier chooses not to disclose a customer when sales drop below this

⁶The results are robust to clustering at the customer–supplier relationship level as well as double clustering at both the customer and supplier level.

10% threshold, we observe a year with a missing sales value and set sales to 0. Doing so is potentially problematic if, for a significant portion of our sample, customer CEO turnovers cause supplier sales to drop just below 10% and suppliers choose not to report those sales. For instance, if a turnover pushes sales from 10% to 9% and we set the missing 9% value to 0%, we overstate the applicable loss in supplier sales, which could bias us towards finding a negative relation between CEO turnovers and supplier sales to customers. This issue would be especially problematic if a nontrivial fraction of suppliers report sales to customers that are just above 10% of their total sales.

To provide insights on the extent of this issue, we tabulate the distributions of PCT_SALES for the full sample and both treatment and control groups in Graphs A and B of Figure 1, respectively. The pattern evident in both graphs suggests that a nontrivial number of suppliers choose not to report sales to a customer when sales account for less than 10% of revenues. As a result, we recognize the need to carefully test the sensitivity of our findings to how we treat missing sales values. In Table 5, we show that our results are robust to addressing this concern from several angles, including dropping all observations with missing sales values between the first and last year of each relationship as well as setting such missing values to a conservative 10% (among other tests). Thus, this concern does not appear to materially affect our main findings.

A second potential concern is that, if the relationship between a supplier and a customer tends to end immediately after the customer replaces its CEO, the addition of 2 years of 0 sales at the end of the relationship may overstate the drop in sales following the turnover event. This overstatement could also bias us towards finding a negative relation between CEO turnovers and supplier sales to customers. To investigate the extent to which this may affect our results, in Table 2, we present descriptive statistics for the distribution of supplier sales to customers for the full, treatment, and control group samples. For the full sample, of the 22,797 customer–supplier years, 5,177 (22.7%) observations stem from the addition of 2 years of 0 sales at the end of the customer–supplier relationship. Of the remaining 17,620 observations that represent the years between the first and last year of the relationship, there are 12,611 years when the supplier reports the customer accounting for 10% or more of revenues, 3,045 years when the supplier reports the customer accounting for between 0% to 10% of revenues, and 1,964 years when the supplier does not report sales to the customer and, thus, the customer accounts for 0% of revenues.⁷ We add 2,337 (17.6%) customer–supplier years of 0 sales to the end of relationships in our treatment group. In contrast, we add 2,840 (29.8%) customer–supplier years of 0 sales to the end of relationships in our control group.

Overall, because we add more years of 0 sales to the end of relationships in the control group than in the treatment group, one could conclude that we are more likely to overstate declines in sales for the control group, which would bias

⁷Ellis, Fee, and Thomas (2012) also find that a substantial fraction of all customers that suppliers report in the segments file actually account for less than 10% of the supplier's revenues. Specifically, they show in Panel C of Table 2 that 34.6% [$55,777 / (55,777 + 104,251)$] of such customers account for less than 10% of the supplier's total sales.

FIGURE 1
Distribution of Supplier Sales to Customers for the Full Sample,
Treatment Group, and Control Group

Figure 1 shows the distribution of PCT_SALES to customers for the full sample, treatment group, and control group. PCT_SALES is sales to a customer divided by the supplier's total sales. The sample includes public suppliers and customers identified from Compustat and ExecuComp over 1993–2011. There are 22,797 customer–supplier years in the full sample. The treatment group consists of customer–supplier pairs in which the customer replaces its CEO at some point during the relationship. The control group consists of customer–supplier pairs in which the customer never replaces its CEO during the relationship. There are 13,274 customer–supplier observations in the treatment group and 9,523 customer–supplier observations in the control group. Graph A tabulates the percentage of total customer–supplier years with respect to the full sample that is in each range. Graph B tabulates the percentage of total customer–supplier years with respect to the treatment or control group that is in each range.

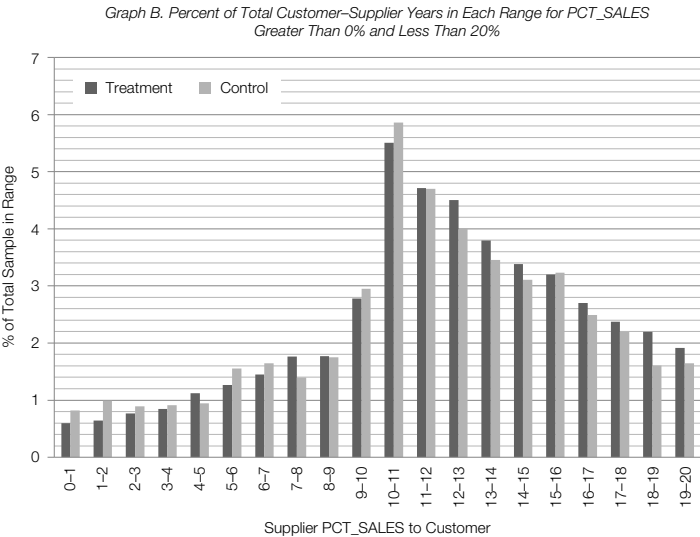
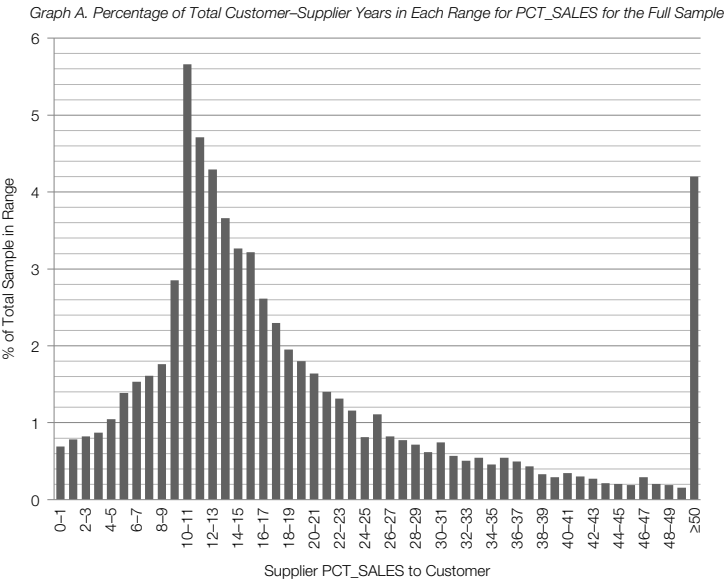


TABLE 2
Distribution of Supplier Sales to Customers

Table 2 reports descriptive statistics for the distribution of supplier sales to customers for the full sample, treatment group, and control group. The sample includes public suppliers and customers identified from Compustat and ExecuComp over 1993–2011. The treatment group consists of customer–supplier pairs in which the customer replaces its CEO at some point during the relationship. The control group consists of customer–supplier pairs in which the customer never replaces its CEO during the relationship. There are 5,876 unique customer–supplier pairs, 2,586 unique customer–supplier pairs in the treatment group and 3,290 unique customer–supplier pairs in the control group.

No. of Customer–Supplier Years...	Full Sample		Treatment Group		Control Group	
	No. of Obs.	Percent of Total Group Obs.	No. of Obs.	Percent of Total Group Obs.	No. of Obs.	Percent of Total Group Obs.
When a customer accounts for at least 10% of its supplier's total sales	12,611	55.3%	7,616	57.4%	4,995	52.5%
When a customer accounts for more than 0% but less than 10% of its supplier's total sales	3,045	13.4%	1,725	13.0%	1,320	13.9%
When a customer accounts for 0% of its supplier's total sales (missing values between first and last year of customer–supplier relationships)	1,964	8.6%	1,596	12.0%	368	3.9%
Due to the addition of 2 years of 0 sales at the end of the customer–supplier relationship to account for the relationship deteriorating rather than just disappearing	5,177	22.7%	2,337	17.6%	2,840	29.8%
Total customer–supplier years	22,797	100.0%	13,274	100.0%	9,523	100.0%

us against finding that supplier sales drop following customer CEO turnovers. Nevertheless, we find similar results when using other methods to capture the deterioration of customer–supplier relationships, including i) adding only 1 year of 0 sales to the end of each relationship and ii) modeling the likelihood that a relationship ends rather than examining changes in sales. We report these results in Table 5 and further discuss these findings in Section IV.A. In sum, this second concern also does not appear to have a material effect on our main findings and conclusions.

IV. Empirical Results

A. CEO Turnovers and Supplier Sales to Customers

Table 3 presents summary statistics and detailed definitions for our variables of interest. Continuous variables are winsorized at their 1st and 99th percentiles each year, and dollar values are expressed in 2009 dollars. Following prior studies (Fee, Hadlock, and Thomas (2006), Banerjee et al. (2008), and Cen, Dasgupta, Elkamhi, and Pungaliya (2016)), we measure supplier sales to customers as PCT_SALES, which is defined as supplier sales to a customer scaled by the supplier's total sales (in percent). The mean (median) of PCT_SALES is 13.7% (11.0%). In our regressions, we include as control variables several supplier characteristics that relate to their dependence on one or more large customers. Banerjee et al. (2008) show that dependent suppliers tend to be smaller, have been publicly traded for fewer years, have more growth and investment opportunities, and have lower financial leverage ratios. Therefore, we include $\ln(\text{SALES})$, $\ln(\text{FIRM_AGE})$, SALES_GROWTH , $\text{R\&D}$, CAPEX , and LEVERAGE as control variables. These dependent suppliers also tend to be more profitable

TABLE 3
Summary Statistics

Table 3 reports descriptive statistics for our variables of interest and control variables. The sample includes public suppliers and customers identified from Compustat and ExecuComp over 1993–2011 and 22,797 customer–supplier years. Continuous variables are winsorized at their 1st and 99th percentiles each year, and dollar values are expressed in 2009 dollars. Variable definitions in parentheses refer to Compustat designations where appropriate. PCT_SALES is sales to a customer divided by the supplier's total sales (*sale*) (in percent). REL_LENGTH is the number of years since a supplier first reported positive sales to a particular customer. ROA is operating income before depreciation (*oidbp*) divided by book value of assets (*at*). SALES_GROWTH is the percentage change in sales over the prior year ($sale_t / sale_{t-1} - 1$). SALES is total sales (*sale*) in millions. VOLATILITY is the annualized standard deviation of daily stock returns during the fiscal year. FIRM_AGE is the number of years since the firm began trading on CRSP. SG&A is selling, general, and administrative expenses (*xsga*) divided by book value of assets (*at*). R&D is research and development expenditures (*xrd*) divided by book value of assets (*at*). CAPEX is capital expenditures (*capx*) divided by book value of assets (*at*). LEVERAGE is the book value of long-term debt (*dltt*) plus debt in current liabilities (*dltc*) divided by book value of assets (*at*). DIVIDEND_PAYER is an indicator variable set to 1 if a firm pays a common dividend (*dvc*) during a fiscal year, and 0 otherwise. BLOCKHOLDER is an indicator variable set to 1 if the supplier has a beneficial owner of more than 10% of a class of security and who is not an employee of the firm. Variables labeled with a leading CUST_ indicate the variable is for the customer. CUST_ΔASSETS is the percentage change in book value of a customer's assets over the prior year ($at_t / at_{t-1} - 1$). CUST_STOCK_RETURN is the 3-year buy-and-hold return of the customer in excess of the 3-year buy-and-hold return of the CRSP value-weighted market portfolio (at least 6 months of return data are required).

	Mean	Std. Dev.	P25	Median	P75
<i>Dependent Variable</i>					
PCT_SALES	13.70	16.45	0.00	11.00	18.00
<i>Supplier Control Variables</i>					
REL_LENGTH	3.89	4.52	1.00	2.00	5.00
ROA	0.04	0.23	−0.01	0.10	0.16
SALES_GROWTH	0.23	0.65	−0.05	0.10	0.32
SALES	1,169	3,908	48.74	165.20	604.60
VOLATILITY	0.71	0.38	0.44	0.63	0.88
FIRM_AGE	12.74	13.42	3.00	8.00	17.00
SG&A	0.31	0.28	0.12	0.25	0.42
R&D	0.08	0.13	0.00	0.03	0.12
CAPEX	0.06	0.06	0.02	0.04	0.07
LEVERAGE	0.20	0.22	0.01	0.14	0.32
DIVIDEND_PAYER	0.18	0.39	0.00	0.00	0.00
BLOCKHOLDER	0.10	0.31	0.00	0.00	0.00
<i>Customer Control Variables</i>					
CUST_SALES	64,149	82,208	11,156	31,865	81,194
CUST_ROA	0.15	0.07	0.10	0.15	0.19
CUST_R&D	0.03	0.05	0.00	0.01	0.06
CUST_LEVERAGE	0.25	0.15	0.13	0.24	0.33
CUST_ΔASSETS	0.12	0.29	0.00	0.07	0.15
CUST_STOCK_RETURN	0.17	1.06	−0.38	−0.02	0.42

(Patatoukas (2012)); have lower selling, general, and administrative (SG&A) expenses (Kalwani and Narayandas (1995)); and pay fewer dividends (Wang (2012)). Thus, we also include ROA, SG&A, and DIVIDEND_PAYER as control variables. These findings are generally consistent with suppliers that have a concentrated customer base producing unique products for customers, lowering their risk of financial distress, and realizing efficiencies from coordination and collaboration along the supply chain.

To the extent that customers are more willing to enter into relationships with suppliers that have better monitoring and lower volatility, we also control for the presence of a large blockholder (BLOCKHOLDER) and return volatility (VOLATILITY). We further consider the possibility of a natural decay rate in sales from a dependent supplier to a customer over time by controlling for the number of years that the supplier has been selling to that customer (REL_LENGTH). Last, we include control variables related to the customer firm, including sales ($\ln(\text{CUST_SALES})$), profitability (CUST_ROA), prior stock performance (CUST_STOCK_RETURN), investment in R&D (CUST_R&D), financial leverage (CUST_LEVERAGE), and asset sales (CUST_ΔASSETS).

Table 4 presents the results of the analysis testing our main hypothesis. The dependent variable in models 1–4 is the natural logarithm of 1 plus PCT_SALES. Model 1 includes only the variable POST_TURNOVER along with year and customer–supplier fixed effects and shows that sales to a customer as a fraction of supplier total sales decrease by 11.9% following the replacement of the customer’s CEO.⁸ Using the sample mean of PCT_SALES (13.7%) and the dollar value of total supplier sales (\$1,169 million), this result implies that suppliers lose, on average, \$19.1 million ($0.137 \times 0.119 \times 1,169$) in sales to a customer that replaces its CEO.

Model 2 of Table 4 adds controls for supplier characteristics, and model 3 further controls for customer characteristics.⁹ In model 4, we account for possible industry-level shocks that can affect both the replacement of a customer’s CEO and supplier sales to the customer by adding industry-year fixed effects (2-digit SIC industries). Although including the full set of control variables and controlling for omitted variables related to industry-level shocks reduces the economic significance of POST_TURNOVER, the coefficient estimate remains negative and statistically significant.

Models 5–7 of Table 4 investigate the robustness of our findings to using alternative specifications of the dependent variable. First, because changes in sales to customers affect both the numerator and the denominator in PCT_SALES, we define PCT_ASSETS as sales to a customer divided by the supplier’s total assets (in percent). Second, to proxy for whether customers stop purchasing from their dependent suppliers altogether in a given year (or sales from these suppliers drop below the 10% reporting threshold), we create an indicator variable (SIGN_CUST) set to 1 if a supplier reports positive sales to a customer, and 0 otherwise.¹⁰ Last, model 7 replaces the dependent variable with the natural logarithm of the supplier’s total sales to capture the effect of customer CEO turnovers on the supplier’s aggregate sales. To allow total sales to vary following turnovers, this model excludes the control variables for supplier size and sales growth. Any drop in sales should decrease the supplier’s total sales if it is not recovering lost revenues by selling to a different customer. However, if a supplier is moving its business to a different customer or the observed drop in sales is the result of measurement error associated with the 10% reporting threshold, then we would not expect to find a decline in total supplier sales. Across all three models, we

⁸We compute the economic significance as follows. The decrease in $\ln(1 + \text{PCT_SALES})$ of 0.127 log points corresponds to a decrease in PCT_SALES of $e^{(-0.127)} - 1 = 11.9$ percentage points. We use this method throughout the paper to interpret estimates from regressions with log-dependent variables. Since our dependent variable is the natural logarithm of 1 plus PCT_SALES, the interpretations of semi-log coefficients are approximate estimates.

⁹Incoming CEOs are associated with higher stock return volatility in their first few years in office (Clayton, Hartzell, and Rosenberg (2005), Pan, Wang, and Weisbach (2015)), which may affect their relationships with dependent suppliers through various channels. Therefore, in untabulated results, we examine whether our main results are robust to controlling for 1-year and 3-year customer stock return volatility. We find that our main results continue to hold and the estimated coefficient on customer return volatility is statistically insignificant.

¹⁰For the dependent variables PCT_ASSETS and SIGN_CUST, we tabulate robust results corresponding to the cross-sectional tests in Tables 6 and 7, which we report in Tables A1 and A2 in the Internet Appendix (available at www.jfqa.org).

TABLE 4
CEO Turnovers and Supplier Sales to Customers

Variable	1	2	3	4	5	6	7	8	9	Average across Suppliers for Customer
POST_TURNOVER	-0.127*** (-3.59)	-0.096*** (-2.80)	-0.081** (-2.38)	-0.091*** (-2.61)	-0.065* (-1.93)	-0.043*** (-3.63)	-0.026** (-2.06)	-0.205*** (-5.68)	-0.175*** (-3.43)	-0.165*** (-3.20)
REL_LENGTH		-0.060*** (-4.83)	-0.078*** (-6.34)	-0.265*** (-4.45)	-0.089*** (-6.94)	-0.027*** (-6.31)	0.033*** (6.70)	-0.124*** (-5.86)	-0.083*** (-3.96)	-0.087*** (-5.29)
ROA		0.163 (1.42)	0.130 (1.13)	0.096 (0.83)	0.674*** (5.44)	-0.007 (-0.18)	1.207*** (18.13)	0.027 (0.17)	0.184 (1.16)	0.267
SALES_GROWTH		0.010 (0.49)	0.005 (0.22)	-0.014 (-0.67)	-0.007 (-0.39)	0.007 (0.12)		-0.003 (-0.08)	-0.015 (-0.57)	-0.039 (-0.81)
ln(SALES)		-0.016 (-0.38)	-0.029 (-0.68)	0.022 (0.51)	0.132*** (3.39)	0.012 (0.88)		0.076 (1.32)	-0.088 (-1.54)	-0.081** (-2.33)
VOLATILITY		-0.020 (-0.39)	-0.022 (-0.42)	-0.045 (-0.84)	0.012 (0.23)	0.011 (0.63)	-0.227*** (-9.31)	0.014 (0.20)	-0.024 (-0.36)	-0.030 (-0.32)
ln(FIRM_AGE)		-0.787*** (-11.37)	-0.779*** (-11.33)	-0.720*** (-10.30)	-0.673*** (-10.57)	-0.261*** (-11.50)	0.218*** (8.40)	-0.605*** (-6.54)	-0.858*** (-8.15)	-0.170*** (-3.48)
SG&A		-0.364*** (-3.17)	-0.365*** (-3.20)	-0.365*** (-3.24)	0.466*** (4.31)	-0.081** (-2.17)	0.162*** (2.62)	-0.387*** (-2.59)	-0.331** (-2.16)	-0.387*** (-2.27)
R&D		-0.061 (-0.24)	-0.103 (-0.40)	-0.019 (-0.08)	0.738*** (3.40)	-0.038 (-0.49)	-0.016 (-0.11)	-0.056 (-0.18)	0.158 (0.47)	0.858* (1.92)

(continued on next page)

Table 4 reports results from ordinary least squares (OLS) fixed effects regressions using public suppliers and customers identified from Compustat and ExecuComp over 1993–2011. The dependent variable in models 1–4 and 8–10 is the natural logarithm of 1 plus PCT_SALES. PCT_SALES is sales to a customer divided by the supplier's total sales (in percent). The dependent variables in models 5–7 are as follows. ln(1 + PCT_ASSETS) is the natural logarithm of 1 plus sales to a customer divided by the supplier's total assets (in percent). SIGN_CUST is an indicator variable set to 1 if a supplier reports positive sales to a customer, and 0 otherwise. ln(SALES) is the natural logarithm of a supplier's total sales. Model 8 restricts the sample to observations within ± 3 years surrounding the CEO turnover event. In model 9, we count the number of dependent suppliers reporting a particular customer each year and exclude observations with an above-sample-median number of dependent suppliers. In model 10, using all customer-supplier observations, we calculate the mean of each supplier variable for each customer so that our sample consists of only one customer-supplier observation per year in which the supplier observation represents the average dependent supplier for that customer. POST_TURNOVER is an indicator variable set to 1 for the years after a customer replaces its CEO, and 0 otherwise. Industries are defined at the 2-digit SIC level. Table 3 provides definitions of control variables. Continuous variables are winsorized at their 1st and 99th percentiles each year, and dollar values are expressed in 2009 dollars. Standard errors are corrected for heteroskedasticity and clustering at the supplier level in all models except model 10 in which standard errors are clustered at the customer level (*t*-statistics are reported in parentheses). *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

TABLE 4 (continued)
CEO Turnovers and Supplier Sales to Customers

Variable	1	2	3	4	5	6	7	8	9	Average across Suppliers for Each Customer
CAPEX		0.344 (1.16)	0.222 (0.74)	0.031 (0.10)	0.315 (1.14)	0.064 (0.60)	0.711*** (5.55)	0.303 (0.60)	0.466 (1.16)	0.257 (0.46)
LEVERAGE		-0.296*** (-2.83)	-0.283*** (-2.69)	-0.252** (-2.42)	-0.233** (-2.17)	-0.071** (-2.00)	0.528*** (8.20)	-0.177 (-1.24)	-0.325** (-2.43)	0.024 (0.14)
DIVIDEND_PAYER		0.049 (0.76)	0.048 (0.76)	0.040 (0.60)	0.023 (0.35)	0.008 (0.34)	0.002 (0.07)	-0.018 (-0.22)	0.023 (0.21)	0.369*** (3.35)
BLOCKHOLDER		-0.070* (-1.91)	-0.071* (-1.93)	-0.055 (-1.50)	-0.050 (-1.37)	-0.022* (-1.71)	-0.034** (-2.04)	-0.036 (-0.72)	-0.079 (-1.51)	0.056 (0.66)
ln(CUST_SALES)			0.396*** (5.74)	0.479*** (6.70)	0.411*** (6.23)	0.114*** (5.02)	0.126*** (4.15)	0.621*** (6.64)	0.249** (2.52)	0.238*** (3.67)
CUST_ROA			0.836** (2.52)	0.925*** (2.72)	0.933*** (2.99)	0.236** (2.11)	-0.056 (-0.43)	0.673 (1.60)	0.796* (1.91)	0.409 (1.04)
CUST_R&D			0.741 (0.80)	0.646 (0.68)	0.681 (0.79)	0.204 (0.65)	-1.539*** (-3.79)	0.369 (0.29)	0.554 (0.48)	-0.241 (-0.23)
CUST_LEVERAGE			0.303 (1.48)	0.114 (0.54)	0.397** (2.04)	0.113 (1.60)	-0.044 (-0.63)	0.590** (2.14)	0.207 (0.76)	0.246 (0.97)
CUST_ASSETS			0.093** (2.42)	0.088** (2.25)	0.095*** (2.58)	0.035*** (2.58)	-0.003 (-0.21)	0.051 (0.94)	0.127** (2.23)	0.280*** (4.99)
CUST_STOCK_RETURN			0.029** (2.28)	0.029** (2.33)	0.023* (1.89)	0.010** (2.25)	0.005 (0.91)	0.041** (2.23)	0.020 (1.29)	0.006 (0.33)
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Customer-supplier FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
Industry-year FEs	No	No	No	Yes	No	No	No	No	No	No
Customer FEs	No	No	No	No	No	No	No	No	No	Yes
No. of obs.	22,797	22,797	22,797	22,797	22,797	22,797	22,797	10,435	11,343	5,071
Adj. R ²	0.385	0.413	0.414	0.456	0.470	0.302	0.964	0.466	0.413	0.318

continue to find a negative relation between customer CEO turnovers and supplier sales.

Next, in model 8 of Table 4, we address the concern that the pre and post periods surrounding the turnover event are too long to attribute changes in supplier sales to the turnover. Specifically, we restrict our sample to the ± 3 years surrounding a turnover and include only customer–supplier observations for customers that replace their CEOs. The coefficient estimate on POST_TURNOVER remains negative and statistically significant. In models 9 and 10, we address the concern that CEO departures at customer firms that have many dependent suppliers (e.g., Walmart) drive our findings. In model 9, we count the number of dependent suppliers reporting a particular customer each year and exclude observations with above sample median values. In model 10, using all customer–supplier observations, we calculate the mean of each supplier variable for each customer-year. This method adjusts our sample to include only one customer–supplier observation per year, where each supplier observation represents the average dependent supplier for that customer. Overall, the coefficient estimate on POST_TURNOVER remains negative and statistically significant in both models.

Models 1 and 2 of Table 5 examine the extent to which our selected approach of matching customer–supplier relationships (i.e., matching from the first year of the relationship to 2 years after it ends) influences our findings. Model 1 shows that our results hold after matching each customer to its suppliers from the first year of the relationship to 1 year after the relationship ends, setting sales to 0 in that year. Similarly, model 2 shows that our results hold after matching each customer to its suppliers beginning 2 years before the first year of the relationship to 2 years after the relationship ends, setting sales to 0 in the 2 years before and after the relationship.¹¹

In models 3–6 of Table 5, we examine the sensitivity of our findings to varying how we treat the years when sales to a customer are missing between the first and last year of the relationship. Because we cannot distinguish whether these missing values are truly 0 or some value below the 10% reporting requirement, in models 3 and 4, we exclude them.¹² In models 5 and 6, we apply a more conservative approach and set such missing values to 10%. Models 3 and 5

¹¹ Our results are also robust to adding 1 year of 0 sales to the beginning of each customer–supplier relationship and extending each relationship to 1 year after the relationship ends, setting sales to 0 in the final year.

¹² We also examine whether our results are robust to excluding all observations with missing sales values between the first and last year of each customer–supplier relationship and observations added to the end of each relationship. Using this sample, the results show an insignificant relation between a supplier's sales to a customer and the replacement of the customer's CEO. However, these results are not entirely surprising because we can only identify a relationship year if a supplier reports sales to a customer, which typically occurs when sales to the customer are 10% or more. Thus, by not capturing any deterioration in the relationship (i.e., sales going to 0) in the final year of the relationship, we significantly bias ourselves from finding an effect of customer CEO turnovers on supplier sales. Overall, this result along with the other findings in Table 5 suggests that, while it is important for our models to capture deterioration in each relationship by adding at least 1 year of 0 sales to the end of each relationship, how we treat observations with missing sales values between the first and last year of each relationship has little effect on our findings.

TABLE 5
CEO Turnovers and Supplier Sales to Customers: Robustness to Assumptions for Treating Years with 0 Sales

Table 5 reports results from OLS fixed effects regressions using public suppliers and customers identified from Compustat and ExecuComp over 1993–2011. The dependent variable in models 1–6 is the natural logarithm of 1 plus PCT_SALES. PCT_SALES is sales to a customer divided by the supplier's total sales (in percent). Model 1 uses the sample created by matching each ExecuComp customer to its suppliers from the first year of the customer–supplier relationship (CSR) to 1 year after the relationship ends. Model 2 uses the sample created by matching each ExecuComp customer to its suppliers from the 2 years before the CSR to 2 years after the relationship ends. PCT_SALES takes a value of 0 in the 2 years before the start of each CSR. Models 3 and 4 exclude all observations between the first and last year of the CSR in which PCT_SALES equals 0%, while models 5 and 6 set PCT_SALES to 10% if it equals 0% between the first and last year of the CSR. Note that models 4 and 6 also use the sample created by matching each ExecuComp customer to its suppliers from the first year of the CSR to 1 year after the relationship ends. In models 7 and 8, the dependent variable is an indicator variable set to 1 in the last year of a CSR, and 0 otherwise. Models 9 and 10 report the hazard ratios from a Cox proportional hazard model analyzing the hazard of a CSR ending. For the analyses in models 7–10, we restrict the sample to observations between the first and last year of a CSR. Models 8 and 10 also exclude all observations between the first and last year of the CSR in which PCT_SALES equals 0%. POST_TURNOVER is an indicator variable set to 1 for the years after a customer replaces its CEO, and 0 otherwise. Every model specification includes the full set of control variables from model 3 of Table 4. Table 3 provides definitions of control variables. Continuous variables are winsorized at their 1st and 99th percentiles each year, and dollar values are expressed in 2009 dollars. Standard errors are corrected for heteroskedasticity and clustering at the supplier level (<i>t</i> -statistics are reported in parentheses). *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.										
	Add Only 1 Year of 0 Sales to the End of Each CSR	Add 2 Years of 0 Sales to the Beginning of Each CSR	Drop All Obs. if PCT_SALES = 0% (model 4 adds only 1 year of 0 sales to the end of each CSR)	Set PCT_SALES = 10% if PCT_SALES = 0% (model 6 adds only 1 year of 0 sales to the end of each CSR)			Probability Relationship Ends (model 8 drops all obs. if PCT_SALES = 0%)		Hazard Relationship Ends (model 10 drops all obs. if PCT_SALES = 0%)	
	1	2	3	4	5	6	7	8	9	10
POST_TURNOVER	−0.068** (−1.96)	−0.128*** (−4.05)	−0.078*** (−2.56)	−0.070** (−2.39)	−0.071** (−2.52)	−0.062** (−2.38)	0.032*** (2.97)	0.032*** (2.95)	1.285*** (5.44)	1.285*** (5.58)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Customer–Supplier FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Supplier FEs	No	No	No	No	No	No	No	No	Yes	Yes
No. of obs.	20,425	31,261	20,833	18,461	22,797	20,425	17,620	15,656	17,620	15,656
Adj. (pseudo-) <i>R</i> ²	0.334	0.507	0.512	0.398	0.500	0.391	0.421	0.413	0.107	0.106

(4 and 6) use the sample in which we match each customer to its suppliers beginning the first year of the relationship to 2 years (1 year) after the relationship ends. In all four models, the results continue to show that suppliers experience declines in sales to customers that replace their CEOs.

In models 7 and 8 of Table 5, we bypass assumptions regarding how to treat missing sales values and how many years to add to the end of each customer–supplier relationship by examining how a turnover event changes the base rate that a relationship ends. To conduct this test, we restrict the sample to observations between the first and last year of each customer–supplier relationship and define the dependent variable as an indicator variable set to 1 in the last year of each relationship. While model 7 keeps observations between the first and last year of each relationship with missing sales values, model 8 drops such observations.¹³ Our results show that customer–supplier relationships are 3.2%–3.7% more likely to end following the replacement of a customer’s CEO. Given that the probability of a relationship ending in any given year is 30.9% (using the sample in model 8), the coefficient estimate on POST_TURNOVER of 0.037 suggests that a customer CEO turnover increases the likelihood of a relationship ending by 12.0% (0.037/0.309). Models 9 and 10 show that the results in models 7 and 8 are robust to estimating a Cox proportional hazard model and imply that the hazard of a customer–supplier relationship ending is 28.5% higher following the replacement of a customer’s CEO.^{14,15}

In sum, the results in Tables 4 and 5 are consistent with the idea that, on average, the replacement of a customer’s CEO disrupts its relationships with dependent suppliers. Moreover, these results are not particularly sensitive to varying how we account for deteriorations in relationships, treat years when sales to a customer are missing, or capture sales to a customer.

B. The Effect of Customer Switching Costs and RSIs

Two key features of customer–supplier relationships that can affect the disruptiveness of customer CEO turnovers on supplier sales are the degree to which a customer finds it costly to switch suppliers and the extent of a supplier’s RSIs. The effect of switching costs on the negative relation between supplier sales and

¹³For the analyses in models 1–8 of Table 5, we tabulate the results corresponding to the cross-sectional tests in Tables 6 and 7 in Tables A3–A10 of the Internet Appendix. The results resemble those reported in Tables 6 and 7.

¹⁴In these tests, we replace the customer–supplier fixed effects with supplier fixed effects due to econometric problems that prohibit estimation of hazard models with fixed effects and nonrepeating events (i.e., because each relationship can only end once, there is only one event per group, resulting in nonrepeated events). See Allison and Christakis (2006) for a discussion of the econometric issues.

¹⁵As a last method to address concerns related to missing sales values and the 10% reporting threshold, we replace PCT.SALES with two measures. First, AVG.PCT.SALES is the average of the previous year, current year, and next year’s values of PCT.SALES in years when PCT.SALES equals 0. Also, if a drop in sales below the 10% threshold is temporary, then next period’s sales should be a good proxy for the continuation of the current customer–supplier relationship. Thus, for the second measure (ADJ.PCT.SALES), we use the next year’s values of PCT.SALES in years when PCT.SALES equals 0. Using these two measures as the dependent variable and the model specification from model 3 of Table 4, the estimated coefficients on POST_TURNOVER are -0.048 (t -statistic = -1.68) and -0.075 (t -statistic = -2.20), respectively. For these two dependent variables, we tabulate the results corresponding to the cross-sectional tests in Tables 6 and 7 in Tables A11 and A12 of the Internet Appendix, which shows similar results.

CEO turnovers is straightforward in that greater switching costs should attenuate the negative relation between CEO turnovers and supplier sales.¹⁶

The effect of RSIs on the negative relation between supplier sales and customer CEO turnovers, however, is unclear. RSIs typically involve unique or specialized products that offer little value outside of these relationships (Titman and Wessels (1988)). When a supplier depends on a customer for a large portion of its revenues, the relationship is often bilateral in that customer investment is also likely required (Banerjee et al. (2008)). Examples of such expenditures include investments in training suppliers or modifications to the production process (Cen, Dasgupta, and Sen (2016)). Thus, these investments can increase the customer's cost of switching suppliers, which would mitigate the decline in supplier sales to a customer that replaces its CEO. However, if a new CEO divests a division that a supplier provides unique products to or changes the firm's product lines, the supplier may not have the production capabilities and flexibility to begin producing alternative goods that fit the customer's new demands. Due to this production inflexibility, the negative relation between CEO turnovers and supplier sales could be more pronounced when suppliers make RSIs.

Table 6 presents the results examining the cross-sectional effect of customer switching costs (Panel A) and RSIs (Panel B) on the negative relation between customer CEO turnovers and supplier sales. We consider three situations when customers face higher switching costs. First, customers will be more dependent on a supplier and, hence, face higher switching costs when they purchase a larger portion of their total inputs from the supplier (Porter (1980), Hui et al. (2012)). Second, customers can find it more difficult to switch suppliers when they have been purchasing from a particular supplier for many years since they are more likely to require products that are not easily substitutable (Brown et al. (2009)). Third, customers face higher switching costs when a supplier captures a larger share of industry sales because there are fewer alternative suppliers that customers can purchase from (Schumacher (1991), Snyder (1996), Inderst and Wey (2007), and Hui et al. (2012)).

We construct four measures to capture instances of higher customer switching costs: i) PCT_COGS is supplier sales to a customer divided by the customer's cost of goods sold, ii) REL_LENGTH is the number of years since the supplier first reported positive sales to a particular customer,¹⁷ iii) SUPPLIER_SIZE is the natural logarithm of the supplier's total sales, and

¹⁶Switching costs not only makes it more difficult for customers to switch suppliers but also inhibits a new CEO's ability to obtain better terms of sales. Suppliers have greater bargaining power and, hence, greater control over negotiations when customers face higher switching costs, which can limit a new CEO's ability to bargain for better terms. Therefore, we expect that the negative relation between customer CEO turnovers and sales from suppliers will also be less pronounced when customers face higher switching costs because new CEOs cannot obtain more favorable terms of sales.

¹⁷In the customer-supplier literature, relationship length is also used as a proxy for RSIs (Banerjee et al. (2007), Brown et al. (2009)) because it is more likely that the relationship involves RSIs if the customer has been purchasing from a supplier for many years. Therefore, a longer relationship can magnify the decline in supplier sales following customer CEO turnovers because suppliers do not have the flexibility to change production. However, because the customer has made RSIs, a longer relationship can also increase its switching costs. Our result that a longer relationship mitigates the negative relation between supplier sales and customer CEO turnovers suggests that relationship length is more likely capturing switching costs in our setting, but there may be alternative interpretations.

TABLE 6
The Effect of Customer Switching Costs and Relationship-Specific Investments

Table 6 reports results from OLS fixed effects regressions using public suppliers and customers identified from Compustat and ExecuComp over 1993–2011. Variable definitions in parentheses refer to Compustat designations where appropriate. The dependent variable in Panels A and B is the natural logarithm of 1 plus PCT_SALES. PCT_SALES is sales to a customer divided by the supplier's total sales (in percent). POST_TURNOVER is an indicator variable set to 1 for the years after a customer replaces its CEO, and 0 otherwise. Customer switching cost measures include PCT_COGS, REL_LENGTH, SUPPLIER_SIZE, and MARKET_SHARE. PCT_COGS equals sales to a customer divided by the customer's cost of goods sold (*cogs*). REL_LENGTH is the number of years since a supplier first reported positive sales to a particular customer. SUPPLIER_SIZE is the natural logarithm of supplier total sales (*sale*) in millions. MARKET_SHARE is supplier sales (*sale*) divided by total industry sales based on 6-digit NAICS industries from the U.S. Economic Census for the supplier. HIGH_SWITCHING_COSTS is an indicator variable set to 1 if the particular customer switching cost measure is above the sample median, and 0 otherwise. R&D is R&D expenditures divided by book value of assets. HIGH_SUPPLIER_R&D and HIGH_CUST_R&D are indicator variables set to 1 if the particular measure is above the sample median, and 0 otherwise. Every model specification includes the full set of control variables from model 3 of Table 4. Table 3 provides definitions of control variables. Continuous variables are winsorized at their 1st and 99th percentiles each year, and dollar values are expressed in 2009 dollars. Standard errors are corrected for heteroskedasticity and clustering at the supplier level (*t*-statistics are reported in parentheses). *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A. The Effect of Customer Switching Costs

	Customer Switching Cost Measure			
	PCT_COGS 1	REL_LENGTH 2	SUPPLIER_SIZE 3	MARKET_SHARE 4
POST_TURNOVER	−0.088*** (−2.77)	−0.522*** (−12.35)	−0.274*** (−5.45)	−0.275*** (−5.21)
POST_TURNOVER × HIGH_SWITCHING_COSTS	0.200*** (5.52)	0.674*** (13.01)	0.353*** (5.45)	0.330*** (4.83)
HIGH_SWITCHING_COSTS	2.077*** (81.86)	−0.465*** (−12.41)	−0.104 (−1.60)	−0.081 (−1.10)
Controls	Yes	Yes	Yes	Yes
Year and customer–supplier FEs	Yes	Yes	Yes	Yes
No. of obs.	22,797	22,797	22,797	21,527
Adj. <i>R</i> ²	0.716	0.426	0.416	0.415

Panel B. The Effect of Relationship-Specific Investments

	1	2	3
POST_TURNOVER	0.078* (1.67)	0.072 (1.50)	0.160*** (2.96)
POST_TURNOVER × HIGH_SUPPLIER_R&D	−0.313*** (−4.86)		−0.253*** (−3.85)
HIGH_SUPPLIER_R&D	0.250*** (3.34)		0.225*** (2.98)
POST TURNOVER × HIGH_CUST_R&D		−0.295*** (−4.32)	−0.215*** (−3.08)
HIGH_CUST_R&D		0.312*** (3.24)	0.288*** (2.93)
Controls	Yes	Yes	Yes
Year and customer–supplier FEs	Yes	Yes	Yes
No. of obs.	22,797	22,797	22,797
Adj. <i>R</i> ²	0.416	0.415	0.416

iv) MARKET.SHARE is the fraction of industry sales captured by the supplier in a given year (based on 6-digit North American Industry Classification System (NAICS) industry sales from the U.S. Economic Census).¹⁸ To indicate

¹⁸Following prior work, we assume that the U.S. Census data are valid for windows of years surrounding a U.S. Economic Census year (Campello (2006), Haushalter, Klasa, and Maxwell (2007), and Klasa, Maxwell, and Ortiz-Molina (2009)). Since our sample ranges from 1993 to 2011 and census years using the NAICS industries are available for 1997, 2002, and 2007, we assume that the values of the 1997, 2002, and 2007 industry sales measures are valid for the 1993–1999, 2000–2004, and 2005–2011 periods, respectively.

relationships in which customers face higher switching costs, we create an indicator variable set to 1 if the particular measure is above the sample median, and 0 otherwise. We then interact these indicator variables with our POST_TURNOVER identifier to test our prediction.

Our results in Panel A of Table 6 show that the decline in $\ln(1 + \text{PCT_SALES})$ to a customer that replaces its CEO is less pronounced when the customer faces higher switching costs. The coefficient estimates on POST_TURNOVER in models 1–4 imply that supplier sales to a customer that replaces its CEO decline by 8.4%–40.7% when the customer faces below-median switching costs.¹⁹ In contrast, the interaction terms in each model imply that suppliers benefit from customer CEO turnovers when customers have greater switching costs, as evidenced by net increases in sales of 5.7% to 16.4%. Using a Wald test, we find that, when customer switching costs are measured by PCT_COGS, REL_LENGTH, and SUPPLIER_SIZE, suppliers experience a statistically positive increase in sales to customers following a turnover event when customers face above-median switching costs (F -statistics of 18.9, 14.4, and 3.3, respectively).

Allen and Phillips (2000) and Kale and Shahrur (2007) suggest that R&D intensive firms are more likely to make RSIs. Furthermore, R&D intensive customers tend to require specialized inputs from suppliers (Levy (1985), Raman and Shahrur (2008)). As such, higher levels of both supplier and customer R&D investment are more likely to indicate instances when a supplier has made RSIs. We define SUPPLIER_R&D as supplier R&D expenses divided by book value of assets and CUST_R&D as customer R&D expenses divided by book value of assets. To indicate instances when suppliers are more likely to have made RSIs, we create an indicator variable set to 1 if a particular measure is above the sample median, and 0 otherwise. We then interact these indicator variables with our POST_TURNOVER identifier to test the effect of RSIs.

The results in Panel B of Table 6 show that customer CEO turnovers benefit suppliers that are less likely to have made RSIs but hurt those that are more likely to have made RSIs. For example, the coefficient estimate on POST_TURNOVER in model 1 implies that supplier sales to a customer that replaces its CEO increase by 8.1% when suppliers are less likely to have made RSIs. However, the coefficient on the interaction term implies that sales decrease by 20.9% when suppliers are more likely to have made RSIs. We find similar results when we measure the likelihood that a supplier made RSIs using customer R&D intensity (model 2) and when we include supplier R&D intensity and customer R&D intensity in the same model (model 3). Using a Wald test, we find that, in models 1 and 2, suppliers experience a statistically significant decrease in sales to customers following a turnover event when suppliers are more likely to have made RSIs (F -statistics of 25.6 and 21.5, respectively). The turnover variable and its interactions with measures of RSI are not significantly different in model 3.

¹⁹We compute the economic significance of interaction terms as follows. The change in $\ln(1 + \text{PCT_SALES})$ following a CEO turnover for a supplier that provides its customer with a larger fraction of the customer's inputs is $0.112 (-0.088 + 0.200)$ in model 1 of Table 6. This increase corresponds to an increase in PCT_SALES of $e^{(-0.088+0.220)} - 1 = 11.9$ percentage points. We use this method throughout the paper to interpret estimates from regressions with log-dependent variables.

Overall, the results of Table 6 suggest that, while the average customer CEO turnover disrupts relationships with dependent suppliers, some suppliers are less affected and may even benefit from the CEO departure. In particular, customer CEO turnovers hurt suppliers when the customer faces lower switching costs and when suppliers are more likely to have made RSIs. However, these events benefit suppliers that are less likely to have made RSIs and whose customers face higher switching costs.

C. The Effect of Outside Successions, Forced Turnovers, and Entrenchment

In this section, we investigate whether the decline in supplier sales following customer CEO turnovers arises from successors breaking up managerial-motivated strategies and investments. We present the results in Table 7. In Panel A, we examine the cross-sectional effect of the type of turnover. Boards often hire outside CEOs to break with the failed policies of incumbents, and forced departures often result from incumbents not acting in the best interests of shareholders. Thus, we expect to find larger declines in supplier sales to customers that replace their CEOs when predecessors are replaced by an outside CEO or forced out of office. Alternatively, outside appointments and forced departures may arise from the need to implement changes and pursue new directions that are not within the expertise of the firm's incumbent CEO.²⁰ Hence, to more directly test whether successors abandon managerial-motivated policies, in Panel B, we examine whether declines in supplier sales following customer CEO turnovers are larger when incumbents are more entrenched.

We define *OUTSIDER* as an indicator variable set to 1 if the successor is an employee of the customer firm for less than 1 year before becoming CEO. To identify forced departures (*FORCED*), we use LexisNexis to search news articles and determine whether the customer forced the incumbent CEO out of office. Following Parrino (1997), we identify a turnover as forced if i) the news article specifically states that the customer forced the CEO out or ii) the departing CEO is less than 60 years old and the departure is not due to poor health, death, or acceptance of a new position. If none of these criteria are met, we classify the turnover as voluntary. Of the 597 turnovers, a customer hires an outsider 181 times (30.3% of all appointments) and forces out the incumbent CEO 170 times (28.5% of all appointments).²¹ We then interact these indicator variables with our *POST_TURNOVER* identifier. We exclude the main effects for *OUTSIDER* and *FORCED* because the interaction terms absorb them.

²⁰For example, Ron Johnson was hired as CEO of J.C. Penney because of his success as a top retail executive at Apple. However, it quickly became evident that following a similar strategy at J.C. Penney (e.g., replacing heavy discount promotions with a relatively standardized pricing model) was not appropriate given J.C. Penney's customer base. As a result, Johnson was forced from his post at J.C. Penney within 2 years of his appointment (Lubin and Mattioli (2013)). In this example, Johnson was fired not due to entrenchment but because his skills did not match what J.C. Penney needed.

²¹Some departures formally announced as voluntary can actually be forced (Jenter and Lewellen (2017)). Therefore, in untabulated results, we identify forced departures as CEO turnovers that follow instances of bottom quartile industry-adjusted (based on 2-digit SIC industries) stock performance each year and find similar results.

TABLE 7
The Effect of Outside Successions, Forced Turnovers, and Entrenchment

Table 7 reports results from OLS fixed effects regressions using public suppliers and customers identified from Compustat and ExecuComp over 1993–2011. The dependent variable in Panels A and B is the natural logarithm of 1 plus PCT_SALES. PCT_SALES is sales to a customer divided by the supplier's total sales (in percent). POST_TURNOVER is an indicator variable set to 1 for the years after a customer replaces its CEO, and 0 otherwise. INSIDER is an indicator variable set to 1 if a customer replaces its CEO with one of the customer's employees, and 0 otherwise. OUTSIDER is an indicator variable set to 1 if a customer replaces its CEO with someone who has worked for the customer for 1 year or less, and 0 otherwise. VOLUNTARY is an indicator variable set to 1 if a customer's CEO voluntarily leaves, and 0 otherwise. FORCED is an indicator variable set to 1 if a customer forces out its CEO, and 0 otherwise. INSIDER_FORCED is an indicator variable set to 1 if INSIDER and FORCED both equal 1, and 0 otherwise. OUTSIDER_VOLUNTARY is an indicator variable set to 1 if OUTSIDER and VOLUNTARY both equal 1, and 0 otherwise. OUTSIDER_FORCED is an indicator variable set to 1 if OUTSIDER and FORCED both equal 1, and 0 otherwise. HIGH_POWER_INDEX is an indicator variable set to 1 if the power index of Adams, Almeida, and Ferreira (2005) is above the sample median, and 0 otherwise. BUSY_BOARD is an indicator variable set to 1 if the fraction of a board's independent directors holding three or more outside board seats is greater than 50%, and 0 otherwise. LOW_BOARD_OWN is an indicator variable set to 1 if the fraction of the shares owned by the firm's board of directors (excluding the CEO) is below the sample median, and 0 otherwise. All three entrenchment measures are calculated in the year before the turnover event. Every model includes the full set of control variables from model 3 of Table 4. Table 3 provides definitions of control variables. Continuous variables are winsorized at their 1st and 99th percentiles each year, and dollar values are expressed in 2009 dollars. Standard errors are corrected for heteroskedasticity and clustering at the supplier level (*t*-statistics are reported in parentheses). *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A. The Effect of Outside Successions and Forced Turnovers

Variable	ln(1+PCT_SALES)		
	1	2	3
POST_TURNOVER	−0.011 (−0.29)	0.024 (0.60)	0.085* (1.95)
POST_TURNOVER × OUTSIDER	−0.316*** (−4.02)		
POST_TURNOVER × FORCED		−0.388*** (−5.65)	
POST_TURNOVER × INSIDER_FORCED			−0.471*** (−5.81)
POST_TURNOVER × OUTSIDER_VOLUNTARY			−0.394*** (−3.84)
POST_TURNOVER × OUTSIDER_FORCED			−0.416*** (−3.83)
Controls	Yes	Yes	Yes
Year and customer–supplier FEs	Yes	Yes	Yes
No. of obs.	22,797	22,797	22,797
Adj. <i>R</i> ²	0.415	0.415	0.416

Panel B. The Effect of Entrenchment

Variable			
	1	2	3
POST_TURNOVER	0.029 (0.62)	0.043 (0.87)	−0.013 (−0.23)
POST_TURNOVER × HIGH_POWER_INDEX	−0.266*** (−3.85)		
POST_TURNOVER × BUSY_BOARD		−0.386*** (−4.98)	
POST_TURNOVER × LOW_BOARD_OWN			−0.173** (−2.23)
Controls	Yes	Yes	Yes
Year and customer–supplier FEs	Yes	Yes	Yes
No. of obs.	20,320	17,546	17,560
Adj. <i>R</i> ²	0.415	0.409	0.407

The results in model 1 of Panel A in Table 7 show that CEO departures resulting in the hiring of outsiders lead to larger post-turnover declines in sales compared to turnovers when customers appoint insiders. The coefficient on POST_TURNOVER implies that sales to customers that replace their CEOs do not significantly decline when customers hire insiders, but the coefficient on the interaction term implies that sales to customers decline by 27.9% when

customers appoint outsiders. The results in model 2 imply that forced turnovers lead to a larger decline in sales compared to voluntary turnovers. The coefficient on POST_TURNOVER implies that sales do not significantly decline when the previous CEO leaves voluntarily, but the coefficient on the interaction term implies that sales decline by 30.5% when a customer forces an incumbent out. The results of a Wald test show that suppliers experience a statistically significant decrease in sales to customers following outside successions and forced turnovers.

In model 3 of Table 7, we investigate the combined effects of forced turnovers and outside successions. We create indicator variables for each of the four combinations of turnovers and exclude the combination when a CEO leaves voluntarily and the customer hires an insider. We then interact these indicator variables with our CEO turnover identifier. Thus, POST_TURNOVER represents the impact on supplier sales when a CEO leaves voluntarily and the customer hires an insider, and the other three combinations are interpreted as incremental effects. The results show that, except when incumbent CEOs leave voluntarily and are replaced by insiders, sales to customers significantly decline following CEO turnovers. However, the results of Wald tests imply that the effects of the other three types of turnovers do not differ statistically.

CEOs are more likely to be entrenched if they have relatively high discretionary decision-making power over the board or if they are less likely to be effectively monitored by the board. Thus, in Panel B of Table 7, we measure the extent to which an incumbent is entrenched using three proxies. Our first measure captures a CEO's decision-making power (POWER_INDEX) using the power index of Adams et al. (2005). This index ranges from 0 to 3 and is the sum of indicator variables for whether a CEO i) is a founder of the firm, ii) is the only insider on the board, and iii) holds both the chairman and president positions. Next, if board members are busy, they are less likely to monitor the CEO. Thus, our second measure captures board busyness and is an indicator variable set to 1 if at least 50% of the firm's independent directors sit on three or more boards (BUSY_BOARD). Our last measure (BOARD_OWN), the fraction of the firm owned by members of the board of directors (excluding the CEO), captures the board's stake in the firm. We use data from Risk Metrics and calculate all three measures as of the year before the turnover and exclude the main effects because the interaction terms absorb them.

The results in Panel B of Table 7 show that the negative relation between customer CEO turnovers and $\ln(1 + \text{PCT_SALES})$ is more pronounced when the incumbent is more likely to have greater decision-making power over the board and weaker board oversight. For instance, there is a negative relation between customer CEO turnovers and $\ln(1 + \text{PCT_SALES})$ only for firms i) whose incumbent CEOs have an above-median power index (model 1), ii) with a busy board (model 2), or iii) with a below-median director ownership stake (model 3). A Wald test indicates that, in all three models, suppliers experience a statistically significant decrease in sales to customers following successions when the incumbent has greater decision-making power and weaker board monitoring. Overall, the results in Table 7 suggest that turnovers disrupt relationships with suppliers because successors enter office and change strategies devised by entrenched incumbents.

D. CEO Turnovers and Supplier Stock Price Reactions to Turnover Announcements

Another way to test whether turnovers disrupt customer–supplier relationships is to examine supplier stock price reactions to the announcement that a customer firm is replacing its CEO. In particular, if CEO departures disrupt customer–supplier relationships, then supplier shareholders should react negatively to the announcement that a customer is replacing its CEO.

We collect customer CEO turnover announcement dates from press reports using LexisNexis and calculate three measures of supplier CARs over the event window $(-2, +2)$, in which $t=0$ is the announcement date. For our first and second measures (MKT_MODEL_CAR and 3-FACTOR_CAR), we obtain CARs using only the CRSP equal-weighted market returns or the three factors described by Fama and French (1993), where parameters are estimated over the window $(-280, -61)$ relative to the announcement date. We require suppliers to have 180 trading days of available return data over the estimation window and also require positive sales from a supplier to a customer during the turnover year. This step leaves us with 425 turnover announcements and 1,491 customer–supplier observations, in which each turnover announcement maps to approximately 3.5 suppliers. Our last measure, CHAR_ADJ_CAR, is characteristic-adjusted abnormal returns obtained by subtracting the returns on a characteristics-matched portfolio from the supplier's returns. We match suppliers to their benchmark portfolio based on market capitalization and the book-to-market ratio in the most recent quarter before the turnover announcement.

Table 8 presents the results of this analysis. In Panels A–C, we cluster standard errors by CEO turnover event to account for cross-sectional correlation due to different suppliers with the same customer having the same announcement date.²² In Panel A, we examine CARs for all turnover announcements. Over the 5-day window around the turnover announcement date and across our three measures of abnormal returns, suppliers experience negative and statistically significant abnormal returns of 0.61% to 0.76%.

In Panel B of Table 8, we compare supplier CARs across different types of turnovers and measures of incumbent entrenchment. If successors break with the failed policies of incumbents, we expect that supplier announcement CARs should be lower when incumbents are forced out of office, replaced by outsiders, or are more entrenched. The results show that suppliers experience a -1.61% to -1.66% CAR to the announcement of an outside appointment (vs. -0.35% to -0.44% CAR to an inside appointment) and a -1.67% to -1.71% CAR to the announcement of a forced turnover (vs. a -0.18% to -0.41% CAR to a voluntary

²² Another approach that can be used to account for this cross-sectional correlation is to create customer CEO turnover event portfolios (MacKinlay (1997)). This method also addresses the concern that the sample of customers with many dependent suppliers drives our results. To apply this approach, we calculate the mean and median supplier CAR across all suppliers for each customer turnover event, which provides us with a sample that consists of only 1 average supplier CAR per turnover event. Overall, we find similar results using this method, except that the difference in CARs for suppliers with below- and above-median director share ownership is indistinguishable from 0. These results are tabulated in Tables A13 and A14 in the Internet Appendix.

TABLE 8
Supplier Stock Price Reactions to CEO Turnover Announcements

Table 8 reports supplier stock price reactions to the announcement that a customer is replacing its CEO using public suppliers and customers identified from Compustat and ExecuComp over 1993–2011. Panels A and B present univariate results, and Panel C presents multivariate results. We require positive sales from suppliers to customers in the year of a CEO turnover announcement, which results in 425 CEO turnover announcements. Cumulative abnormal returns (CARs) are in percentages and are calculated over the event window $(-2, +2)$, in which $t=0$ is the announcement date. Abnormal returns are calculated using i) a market model (MKT_MODEL_CAR) using CRSP equal-weighted market returns, ii) the Fama–French 3-factor model (3-FACTOR_CAR), and iii) size and book-to-market characteristic adjusted returns (CHAR_ADJ_CAR). The parameters for the market and 3-factor models are estimated over the window $(-280, -61)$ relative to the announcement date. The returns on the benchmark size and book-to-market portfolio are based on matching in the prior quarter. CARs are winsorized at their 1st and 99th percentiles. Standard errors in Panels A–C are clustered by customer CEO turnover event. Panel A reports mean CARs for all CEO turnover announcements. P-values indicate the results of tests for whether mean (Student's *t*-Test) CARs are different than 0. Panel B compares CARs across different types of CEO turnovers and measures of incumbent entrenchment. CEO turnover types and measures of entrenchment are defined in Table 7. Results of tests for whether mean (Student's *t*-Test) CARs are different across CEO turnover types and entrenchment measures are reported next to the mean values for the left-hand-side turnover or entrenchment group. To obtain the clustered standard errors in Panels A and B, we run univariate OLS regressions and cluster by turnover event. In Panel C, *t*-statistics are reported in parentheses. Supplier control variables include REL_LENGTH, ROA, SALES_GROWTH, ln(SALES), VOLATILITY, ln(FIRM_AGE), SG&A, R&D, CAPEX, LEVERAGE, DIVIDEND_PAYER, and BLOCKHOLDER. Customer control variables include ln(CUST_SALES), CUST_ROA, CUST_R&D, CUST_LEVERAGE, CUST_ΔASSETS, and CUST_STOCK_RETURN. Table 3 provides definitions of control variables. Continuous variables are winsorized at their 1st and 99th percentiles each year, and dollar values are expressed in 2009 dollars. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A. Supplier Stock Price Reactions for All Turnovers

Variable	No. of Obs.	Mean	p-Value
MKT_MODEL_CAR	1,491	−0.61**	(0.040)
3-FACTOR_CAR	1,491	−0.70**	(0.011)
CHAR_ADJ_CAR	1,491	−0.76**	(0.011)

Panel B. Supplier Stock Price Reactions by Type of Turnover and Measures of Entrenchment

Variable	Outside Successions		Inside Successions	
	No. of Obs.	Mean	No. of Obs.	Mean
MKT_MODEL_CAR	310	−1.61*	1,181	−0.35
3-FACTOR_CAR	310	−1.66*	1,181	−0.44
CHAR_ADJ_CAR	310	−1.55	1,181	−0.55
Variable	Forced Turnovers		Voluntary Turnovers	
	No. of Obs.	Mean	No. of Obs.	Mean
MKT_MODEL_CAR	418	−1.71**	1,073	−0.18
3-FACTOR_CAR	418	−1.54	1,073	−0.36
CHAR_ADJ_CAR	418	−1.67*	1,073	−0.41
Variable	Outside Successions and Forced Turnovers		Inside Successions and Voluntary Turnovers	
	No. of Obs.	Mean	No. of Obs.	Mean
MKT_MODEL_CAR	155	−2.61***	918	−0.11
3-FACTOR_CAR	155	−2.47**	918	−0.28
CHAR_ADJ_CAR	155	−2.73***	918	−0.41
Variable	Above-Median Power Index		Below-Median Power Index	
	No. of Obs.	Mean	No. of Obs.	Mean
MKT_MODEL_CAR	497	−1.49*	682	−0.22
3-FACTOR_CAR	497	−1.51*	682	−0.43
CHAR_ADJ_CAR	497	−1.51	682	−0.54
Variable	Busy Board of Directors		Not Busy Board of Directors	
	No. of Obs.	Mean	No. of Obs.	Mean
MKT_MODEL_CAR	416	−0.84	602	−1.00
3-FACTOR_CAR	416	−1.00	602	−1.13
CHAR_ADJ_CAR	416	−1.42	602	−0.67
Variable	Below-Median Board Ownership		Above-Median Board Ownership	
	No. of Obs.	Mean	No. of Obs.	Mean
MKT_MODEL_CAR	509	−1.57*	509	−0.31
3-FACTOR_CAR	509	−1.56	509	−0.59
CHAR_ADJ_CAR	509	−1.45	509	−0.78

(continued on next page)

TABLE 8 (continued)
Supplier Stock Price Reactions to CEO Turnover Announcements

Panel C. Multivariate Analysis: Supplier Stock Price Reactions to CEO Turnover Announcements

Variable	MKT_ MODEL_ CAR	3-FACTOR_ CAR	CHAR_ ADJ_CAR	MKT_ MODEL_ CAR	3-FACTOR_ CAR	CHAR_ ADJ_CAR	MKT_ MODEL_ CAR	3-FACTOR_ CAR	CHAR_ ADJ_CAR
	1	2	3	4	5	6	7	8	9
INSIDER_FORCED	-0.837 (-0.92)	-0.524 (-0.60)	-0.566 (-0.67)				-1.662 (-1.59)	-1.731* (-1.79)	-1.556 (-1.54)
OUTSIDER_VOLUNTARY	-0.399 (-0.40)	-0.221 (-0.25)	0.065 (0.07)				-0.603 (-0.52)	-0.514 (-0.47)	-0.281 (-0.27)
OUTSIDER_FORCED	-2.892*** (-3.16)	-2.673*** (-2.94)	-2.459*** (-3.16)				-2.769* (-1.96)	-2.543* (-1.80)	-2.445** (-2.09)
POWER_INDEX				-1.463** (-2.33)	-1.357** (-2.35)	-1.410** (-2.58)	-1.356** (-2.14)	-1.246** (-2.14)	-1.311** (-2.40)
BUSY_BOARD				0.214 (0.25)	0.437 (0.54)	0.526 (0.67)	0.184 (0.22)	0.417 (0.52)	0.502 (0.65)
BOARD_OWEN				-1.744 (-0.99)	-2.359 (-1.47)	-3.216** (-2.22)	-1.604 (-0.97)	-2.243 (-1.47)	-3.072** (-2.17)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	1,491	1,491	1,491	1,018	1,018	1,018	1,018	1,018	1,018
Adj. R ²	0.011	0.014	0.021	0.008	0.012	0.019	0.010	0.016	0.022

turnover). Furthermore, suppliers experience a -2.47% to -2.73% CAR to the announcement of an outside appointment following a forced turnover but only a -0.11% to -0.41% CAR to the announcement of an inside appointment following a voluntary turnover. Supplier CARs are also more negative when an incumbent is more likely to be entrenched. For incumbents with an above-median power index, supplier CARs are -1.49% to -1.51% (vs. -0.22% to -0.43% for a below-median power index). In addition, when customers have below-median board share ownership, suppliers' CARs are -1.57% (vs. -0.31% for above-median board ownership). Differences are significant across subsamples in most models, with the exception that CARs are not statistically different between firms with and without busy boards.

Panel C of Table 8 provides the results from a multivariate analysis examining the relation between supplier CARs and the type of turnover, incumbent decision-making power, and board oversight.²³ The models regress our three measures of CARs on the same set of control variables used in our main model specification (from model 3 of Table 4) and year fixed effects. The results support our univariate findings. Models 1–3 show that CARs to announcements of outside successions following forced turnovers are significantly more negative than CARs to announcements of inside successions following voluntary turnovers. Models 4–6 show a negative relation between supplier CARs and the incumbent's power index. Last, models 7–9 include both turnover types and entrenchment measures in the same model and confirm the results in models 1–6. Relative to inside successions following voluntary turnovers, outside successions following forced

²³In untabulated analyses, we also examine whether supplier CARs vary with the degree to which it is costly for customers to switch suppliers as well as with the likelihood that suppliers have made RSIs. Our measures of customer switching costs and RSIs are uncorrelated with the measures of supplier CARs.

turnovers have CARs that are 2.45% to 2.77% lower. Furthermore, a 1-standard-deviation increase in the incumbent's power index (0.70) is associated with 0.87% to 0.95% lower CARs. Overall, the results in Table 8 are consistent with customer CEO turnovers disrupting their relationships with dependent suppliers.

E. Potential Mechanisms for Why Supplier Sales Decline

Disruptions in relationships with dependent suppliers can arise from successors divesting assets or business segments, reallocating resources across segments, switching suppliers, or renegotiating terms of sales. In this section, we attempt to distinguish between these potential mechanisms. Table 9 presents the results of this analysis.

We use an approach similar to that in Pan et al. (2016). Specifically, we create several measures to capture each mechanism and regress these measures on an indicator variable that is set to 1 if a CEO's tenure is between 0 and 2 years (TURNOVER[0,2]). Controls include CEO age, firm age, industry-adjusted ROA, industry-adjusted stock returns, market-to-book ratio, book leverage, an indicator variable for whether the firm pays a common dividend, the natural logarithm of book assets, and year and customer fixed effects. For this test, the sample includes one firm-year observation for the ExecuComp customers in our main sample. The coefficient estimate on TURNOVER[0,2] is important as it indicates what actions new CEOs take in their first few years in office.

Models 1–3 of Panel A in Table 9 present results testing whether successors are more likely to divest or add new operations in their first few years in office. To capture such situations, we first follow Atanassov and Kim (2009) and create an indicator variable set to 1 if the customer's value of PP&E drops by at least 15% relative to the previous year (DIVEST_PP&E). We also create an indicator variable set to 1 in the year that a customer stops reporting a business segment (DIVEST_SEGMENT) and an indicator variable set to 1 in the year that a customer starts reporting a new business segment (ADD_SEGMENT). The results show that new CEOs are significantly more likely to divest PP&E and business segments.²⁴

Models 4–6 of Panel A in Table 9 present the results from testing whether successors are more likely to switch suppliers in their first few years in office. To capture such situations, we create an indicator variable set to 1 if the customer's purchases from all dependent suppliers is greater than or equal to its total purchases from all dependent suppliers in the prior year (SWITCH_SUPPLIERS), an

²⁴ While a supplier may lose sales to a customer divesting assets, it may gain sales from the firm acquiring the assets. To investigate this possibility, we create a sample of customer–supplier years when the customer has a 1-year drop in PP&E greater than 15% during the customer–supplier relationship in the years following the turnover event. We require that the supplier have a drop in sales to the customer over the years $t - 1$ to t or t to $t + 1$. If we do not observe a drop in sales to the customer over these years, then it is unlikely that the customer is divesting a segment that is purchasing from the supplier. This restriction leaves us with a sample of 302 customer–supplier years. Next, we obtain acquisition data from the Thomson Reuters Securities Data Company (SDC) Mergers and Acquisitions database and identify transactions in the years around the divestiture when one of the supplier's existing (or new) customers purchases assets from the divesting customer. We find five such cases and one case where the supplier purchases a division from the customer. Thus, it seems rare for a supplier to recover lost sales from a customer that divests assets by selling to a different customer that purchases those assets.

TABLE 9
Potential Mechanisms for Why Supplier Sales Decline

Table 9 reports results from OLS fixed effects regressions using public customers identified from Compustat and ExecuComp over 1993–2011. The dependent variables in models 1–7 are as follows. DIVEST_PP&E is an indicator variable set to 1 if net property, plant, and equipment drops at least 15% relative to the prior year, and 0 otherwise. DIVEST_SEGMENT is an indicator variable set to 1 in the year when a customer stops reporting a business segment, and 0 otherwise. ADD_SEGMENT is an indicator variable set to 1 if the customer has added a new business segment, and 0 otherwise. SWITCH_SUPPLIERS is an indicator variable set to 1 if the customer's purchases from all suppliers is greater than or equal to its total purchases from all dependent suppliers in the prior year, and 0 otherwise. ADD_NEW_SUPPLIER is an indicator variable set to 1 in the year that a supplier first discloses the customer as a major customer, and 0 otherwise. DROP_SUPPLIER is an indicator variable set to 1 if the customer has fewer suppliers than in the prior year, and 0 otherwise. GROSS_MARGIN is total sales less cost of goods sold divided by total sales (in percent). TURNOVER[0.2] is an indicator variable set to 1 if the CEO is in their first 3 years at the firm (tenure of 0–2 years). DECR_SALES is an indicator variable set to 1 if average sales (the natural logarithm of sales) to a specific customer in the post-turnover period is less than average sales at this customer during the pre-turnover period, and 0 otherwise. Control variables include the age of the CEO in the fiscal year, industry-adjusted 1-year stock returns, market-to-book ratio, book leverage, an indicator variable for whether the firm pays a common dividend, and the natural logarithm of book assets. Continuous variables are winsorized at their 1st and 99th percentiles each year, and dollar values are expressed in 2009 dollars. Standard errors are corrected for heteroskedasticity and clustering at the customer level (*t*-statistics are reported in parentheses). *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A. Multivariate Regression Results

	DIVEST_ PP&E 1	DIVEST_ SEGMENT 2	ADD_ SEGMENT 3	SWITCH_ SUPPLIERS 4	ADD_NEW_ SUPPLIER 5	DROP_ SUPPLIER 6	GROSS_ MARGIN 7
TURNOVER[0.2]	0.032*** (2.68)	0.042*** (2.73)	0.018 (1.25)	−0.036** (−2.01)	−0.076*** (−4.06)	0.041** (2.42)	−0.020 (−0.07)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year and customer FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	4,954	4,954	4,954	4,954	4,954	4,954	4,954
Adj. R ²	0.223	0.130	0.182	0.052	0.238	0.143	0.914

Panel B. Multivariate Regression Results Conditioning on Suppliers Losing Sales Following the Customer CEO Turnover

	1	2	3	4	5	6	7
TURNOVER[0.2]	−0.014 (−0.86)	0.048** (1.98)	0.008 (0.29)	0.017 (0.49)	−0.075** (−2.28)	−0.013 (−0.47)	−0.030 (−0.06)
TURNOVER[0.2] × DECR_SALES	0.060*** (3.10)	−0.008 (−0.29)	0.014 (0.50)	−0.068* (−1.78)	−0.002 (−0.07)	0.069** (2.29)	0.012 (0.02)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year and customer FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	4,954	4,954	4,954	4,954	4,954	4,954	4,954
Adj. R ²	0.225	0.130	0.182	0.053	0.238	0.144	0.914

indicator variable set to 1 in the year that a supplier first discloses the firm as a customer (ADD_NEW_SUPPLIER), and an indicator variable set to 1 if the customer has fewer dependent suppliers than in the prior year (DROP_SUPPLIER). The results show that new CEOs are less likely to increase purchases from dependent suppliers and have a new supplier disclose them as a major customer. New CEOs are also more likely to decrease the number of dependent suppliers.²⁵

Last, model 7 of Table 9 presents the results from testing whether successors are more likely to negotiate better terms of sales (namely, lower prices) in their first few years in office. To test this relation, we assume that, if a successor improves terms of sales, then cost of goods sold should decrease and gross margins should increase. However, our results show that gross margins are no different in the first few years of a successor's tenure compared to other firm-years. Collectively, the results in Panel A support the notion that declines in supplier sales to customers that replace their CEOs appear to be driven by successors divesting assets and business segments. The results are inconsistent with successors switching suppliers or negotiating lower prices.

The results in Panel A of Table 9 show the strategic and operational changes that all successors make, on average. Some decisions, however, may hurt suppliers more than others. Thus, in Panel B, we attempt to understand which actions result in a decrease in supplier sales to the customer. To do so, we set an indicator variable (DECR_SALES) to 1 if average sales (the natural logarithm of sales) to a specific customer in the post-turnover period is less than average sales to the same customer during the pre-turnover period, and 0 otherwise. We then interact this indicator variable with our TURNOVER[0,2] identifier for each of our seven models from Panel A. The main effect for DECR_SALES is excluded because the interaction term absorbs it. A significant interaction term would indicate the specific changes successors make that result in supplier sales decreases.

The results in Panel B show that successors' actions that ultimately lead to a decrease in dependent supplier sales include divesting PP&E (model 1) as well as switching (model 4) and dropping suppliers (model 6). In sum, the results in Table 9 suggest that we observe a decline in sales because successors divest operations.

Another explanation for our findings is that CEO turnovers disrupt key board appointments that helped to initiate customer-supplier relationships. As such, the decline in supplier sales to customers that replace their CEOs may be attributed

²⁵ Note that data limitations prohibit us from observing a direct reallocation of purchases from one supplier to another. Also, due to the nature of the data, we can observe only suppliers that sell 10% or more of total sales to an individual customer (or suppliers that voluntarily report the customer as important to their business). Thus, we will not observe a reallocation of resources unless i) it is to one of the dependent suppliers, ii) the new business pushes another supplier above the 10% sales threshold, or iii) the new supplier begins to report the customer as important to its business. In an additional attempt to examine whether successors switch suppliers, we count the number of suppliers disclosing sales to a particular customer before and after the CEO departure. The average number of suppliers disclosing sales to a particular customer 1, 2, and 3 years before the turnover is 3.94, 3.78, and 3.77, respectively. In contrast, the average number of suppliers disclosing sales to a particular customer 1, 2, and 3 years after the turnover is 3.46, 3.32, and 3.14, respectively. A *t*-test comparing the average number of suppliers disclosing sales to a particular customer over the 3 years before and after the turnover indicates that this difference is statistically significant at the 10% level. This finding provides further evidence that customers are not simply switching suppliers.

to the loss of key directors, not the loss of the CEO. To investigate this possibility, we select a random sample of 100 customer–supplier years when a customer CEO turnover occurs and manually search proxy statements to identify directors of the supplier that were connected to the customer. We select 50 observations from 1998 and 50 from 2006 to allow for differences in the supply and demand of directors after the Sarbanes–Oxley Act of 2002 (Linck, Netter, and Yang (2009)). We find 12 observations in which a supplier director was employed by the customer firm, but in only seven of these cases was the director employed by the customer within the last 10 years. There are no instances of a supplier director contemporaneously working for the customer or of the supplier ever employing a customer director.²⁶ Furthermore, all supplier directors who were employed by the customer within the last 10 years remained with the supplier for a minimum of 3 years after the turnover or until proxy statements were no longer available through the SEC. These findings suggest that changes in board-level director relationships between customers and suppliers unlikely explain our results.

F. CEO Turnovers and Supplier Performance

If the loss in supplier sales following customer CEO departures is financially significant and suppliers are unable to recover lost sales by selling to other customers, then lower post-turnover supplier sales should lead to deterioration in overall supplier financial performance. We test this implication by examining how a loss in sales to a customer impacts several supplier performance measures and report the results in Table 10. Panel A provides regression results with only year and customer–supplier fixed effects as control variables, and Panel B reports the results from regressions that include the full set of control variables.²⁷ To examine the impact of a decrease in sales on supplier performance, we interact the indicator variable *DECR_SALES* with *POST_TURNOVER*.²⁸ The main effect for *DECR_SALES* is excluded because the interaction term absorbs it.

We first examine how a decrease in sales to a customer that replaces its CEO impacts supplier total sales $\ln(\text{SALES})$. Inconsistent with suppliers recovering lost sales by selling to other customers, the results in model 1 of Panel A (Panel B) of Table 10 imply that a supplier's total sales decrease by 7.8% (3.0%) when it loses sales to a customer that replaces its CEO. A Wald test suggests that the joint significance of the level and interaction terms in both Panels A and B is statistically negative.

Second, we investigate how the decrease in sales to a customer affects the supplier's overall firm value, as measured by *TQ* (market value of assets/book

²⁶These results are not surprising given that Dass, Kini, Nanda, and Onal (2014) report that less than 1.2% of their sample firms have a director who is also a director of a customer or a supplier.

²⁷In the regression models in Table 10, we control for supplier size with the natural logarithm of supplier book assets instead of sales. Since we are interested in how a drop in sales affects supplier performance, holding supplier sales constant by including supplier total sales as a control variable restricts our ability to identify the effect of a decrease in sales to a customer that replaces its CEO. Thus, controlling for book assets instead helps alleviate this issue.

²⁸Of the 2,531 customer–supplier pairs that have a customer that replaces its CEO and for which we have data available for our performance measures, supplier sales to the customer decrease in 1,321 (52.2%) of these relationships.

TABLE 10
CEO Turnovers, Supplier Sales to Customers, and Supplier Financial Performance

Table 10 reports results from OLS fixed effects regressions using public suppliers and customers identified from Compustat and ExecuComp over 1993–2011. Variable definitions in parentheses refer to Compustat designations where appropriate. Panel A reports multivariate results from regressions with only year and customer-supplier fixed effects as control variables. Panel B reports multivariate results from regressions with the full set of control variables. The dependent variables in models 1–10 are as follows. ln(SALES) is the natural logarithm of supplier total sales (sale) in millions. ln(TO) is the natural logarithm of the supplier's market value of assets (book assets (at) plus market value of equity (prcc.f × csho) minus book value of equity (ceq)) divided by book value of assets (at). ROA is income before extraordinary items (lccom) divided by the average beginning and end-of-year book value of assets (at). ROE is income before extraordinary items (lccom) divided by the average beginning and end of year book value of equity (ceq). ASSET_TURN is total sales (sale) divided by the average beginning and end-of-year book value of assets (at). PM is income before extraordinary items (lccom) divided by total sales (sale). OPM is operating income before depreciation (olbop) divided by total sales (sale). NON_OPM is nonoperating income (nopi) divided by total sales (sale). GM is total sales (sale) less cost of goods sold (cogs) divided by total sales (sale). SG&A is selling, general, and administrative expenses (xsga) divided by total sales (sale). POST_TURNOVER is an indicator variable set to 1 for the years after a customer replaces its CEO, and 0 otherwise. DECR_SALES is an indicator variable set to 1 if average sales (the natural logarithm of sales) to a specific customer in the post-turnover period is less than average sales to this customer during the pre-turnover period, and 0 otherwise. Supplier control variables include REL_LENGTH, SALES_GROWTH, ln(ASSETS), VOLATILITY, ln(FIRM_AGE), R&D, CAPEX, LEVERAGE, DIVIDEND_PAYER, and BLOCKHOLDER. However, model 1 excludes SALES_GROWTH. Customer control variables include ln(CUST_SALES), CUST_R&D, CUST_LEVERAGE, CUST_ASSETS, and CUST_STOCK_RETURN. Table 3 provides definitions of control variables. Continuous variables are winsorized at their 1st and 99th percentiles each year, and dollar values are expressed in 2009 dollars. Standard errors are corrected for heteroskedasticity and clustering at the supplier level (t-statistics are reported in parentheses). *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.										
Variables	ln(SALES)	ln(TQ)	ROA	ROE	ASSET_TURN	PM	OPM	NON_OPM	GM	SG&A
Panel A. Multivariate Regression Results Excluding Customer and Supplier Control Variables										
POST_TURNOVER	0.075*** (3.12)	0.015 (0.77)	0.018** (2.49)	0.050* (1.66)	0.046*** (2.70)	0.095*** (2.63)	0.064** (2.31)	−0.006 (−1.52)	0.018 (0.85)	−0.016 (−1.61)
POST_TURNOVER × DECR_SALES	−0.156*** (−5.12)	−0.058** (−2.44)	−0.033*** (−3.80)	−0.085** (−2.46)	−0.045** (−2.15)	−0.229*** (−4.46)	−0.181*** (−4.61)	0.007* (1.92)	−0.076** (−2.54)	0.041*** (2.87)
Controls	No	No	No	No	No	No	No	No	No	No
Year and customer-supplier FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	21,328	21,328	21,328	21,328	21,328	21,328	21,328	21,328	21,328	21,328
Adj. R ²	0.956	0.641	0.613	0.234	0.848	0.615	0.664	0.530	0.672	0.746
Panel B. Multivariate Regression Results Including Customer and Supplier Control Variables										
POST_TURNOVER	0.054*** (3.65)	0.016 (0.87)	0.012** (2.12)	0.041 (1.39)	0.046*** (3.00)	0.098*** (2.72)	0.068** (2.40)	−0.006* (−1.75)	0.022 (1.03)	−0.013 (−1.32)
POST_TURNOVER × DECR_SALES	−0.084*** (−4.57)	−0.046** (−2.05)	−0.012* (−1.69)	−0.046 (−1.38)	−0.045** (−2.52)	−0.209*** (−4.03)	−0.167*** (−4.17)	0.010** (2.45)	−0.074** (−2.35)	0.029** (2.10)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year and customer-supplier FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	21,328	21,328	21,328	21,328	21,328	21,328	21,328	21,328	21,328	21,328
Adj. R ²	0.975	0.679	0.721	0.267	0.873	0.642	0.686	0.555	0.689	0.755

value of assets). The results in model 2 of Panel A (Panel B) in Table 10 imply that a supplier's firm value decreases by 4.2% (3.0%) when it loses sales to a customer that replaces its CEO. A Wald test implies that the joint significance of the level and interaction terms in both Panels A and B is statistically negative.

Models 3–10 of Table 10 examine how a drop in sales to a customer impacts the performance measures examined by Patatoukas (2012). Specifically, the dependent variables in models 3–10 are ROA (income before extraordinary items/average beginning and end-of-year book value of assets), ROE (income before extraordinary items/average beginning and end-of-year book value of equity), ASSET_TURN (total sales/average beginning and end-of-year book value of assets), PM (income before extraordinary items/total sales), OPM (operating income before depreciation/total sales), NON_OPM (nonoperating income/total sales), GM (total sales less cost of goods sold/total sales), and SG&A (SG&A expenses/total sales).

Patatoukas (2012) finds that suppliers with more concentrated customer bases realize operational and selling efficiencies by selling to only a few large customers. The results in models 3–10 of Panel A in Table 10 are generally consistent with this notion. In particular, we find that, relative to instances when suppliers do not lose sales to customers that replace their CEOs, suppliers that experience a drop in sales to customers after the turnover event have lower return on assets, return on equity, asset turnover, profit, operating and gross margins, and higher SG&A expenses. The results in Panel B are similar except that return on equity is not statistically lower. Using a Wald test, we find that the joint significance of the variables is statistically negative for all models in Panel A (Panel B) except for models 4, 5, and 8 (3, 4, 5, and 8). Collectively, the results in Table 10 suggest that a decline in supplier sales to a customer that replaces its CEO leads to lower overall supplier financial performance.

V. Robustness Tests and Endogeneity

A. Endogeneity Concerns

Thus far, we have taken the following steps to alleviate endogeneity concerns. First, we include customer–supplier fixed effects in all models. If omitted variables are time-invariant and unique to each customer–supplier pair, then using a fixed effects estimator will result in consistent estimates. Second, we conduct cross-sectional tests across numerous dimensions. For an omitted variable to explain our results, in addition to being uncorrelated with our included control variables, it would also have to be correlated with all of the characteristics for which we find cross-sectional variation between CEO turnovers and supplier sales to these customers. As such, interaction terms are less prone to an omitted variable bias, as it is much less likely that an omitted variable is correlated with the interaction term than the level term (Raddatz (2006), Lin, Ma, Malatesta, and Xuan (2012)). In the remainder of this section, we discuss the additional tests we conduct to further rule out endogeneity concerns.

To provide added assurance that our results do not suffer from an omitted variable bias, we examine a propensity score matched sample. In all of our models, we control for several supplier and customer characteristics. However, CEO turnovers could be picking up nonlinear effects if the linear controls used

throughout our models do not adequately account for differences between suppliers dependent on customers that replace their CEOs and suppliers dependent on customers that do not. Thus, we use a propensity score matched sample to correct for endogenous selection on observed variables (Rosenbaum and Rubin (1983), Dehejia and Wahba (2002)) and compare turnover samples to nonturnover samples. Using a linear probability model, we first regress an indicator variable set to 1 if a supplier's customer replaces its CEO at some point during the relationship (i.e., in treatment group) on our previously used control variables and estimate the probability (i.e., the propensity score) that a supplier sells to a customer that replaces its CEO.²⁹ Second, we match each observation in the treatment group to an observation in the control group with the closest propensity score. We match without replacement and require the propensity scores for each matched pair to be within $\pm 2.5\%$ of each other.³⁰ The resulting sample consists of 4,167 customer-supplier years for both the treatment and control groups.

Panel A of Table 11 reports univariate results comparing the means and medians of control variables across the treatment and control groups. Based on mean (median) values, the only control variables that are statistically different between samples are REL_LENGTH, CUST_LEVERAGE, and CUST_ΔASSETS (REL_LENGTH, ROA, SALES_GROWTH, CUST_LEVERAGE, and CUST_STOCK_RETURN).

In Panel B of Table 11, we first examine the change in PCT.SALES due to a customer replacing its CEO in a difference-in-differences nonregression setting. For this analysis, we use the matched sample and calculate the difference in PCT.SALES for the treatment suppliers before and after the turnover event minus the difference in PCT.SALES for the control suppliers before and after the turnover event. Therefore, this test uses a total of 694 treatment suppliers and their matched control suppliers for which we have data in both the pre- and post-turnover periods. The results in row 1 show that, relative to control suppliers, PCT.SALES decreases by an additional 5.85 percentage points for treatment suppliers and this result is statistically significant at the 1% level.

A second potential endogeneity concern arises because some turnovers are themselves endogenously determined. For example, we argue that the main mechanism underlying our findings is that a new CEO implements new strategies that harm the firm's existing suppliers. This mechanism implies that it is the CEO who makes these strategic decisions. However, it is also likely that, in some instances, the board of directors recognizes the need for strategic change, replaces the CEO, and hires a new CEO, in part, to implement this change. If our main results are caused by the latter explanation, then they may be spurious because it is a board decision that leads to a disruption in sales rather than the turnover causing the change. To account for this concern, we use the propensity score matched

²⁹Control variables include REL_LENGTH, ROA, SALES_GROWTH, ln(SALES), VOLATILITY, ln(FIRM_AGE), SG&A, R&D, CAPEX, LEVERAGE, DIVIDEND_PAYER, BLOCKHOLDER, ln(CUST_SALES), CUST_ROA, CUST_R&D, CUST_LEVERAGE, CUST_ΔASSETS, CUST_STOCK_RETURN, and industry-year fixed effects. We present first-stage regression results in Table A15 in the Internet Appendix.

³⁰We use the $\pm 2.5\%$ cutoff so that matched firms are very similar. The results are robust to using a $\pm 1.0\%$, $\pm 5.0\%$, or $\pm 10.0\%$ cutoff point.

sample and limit the sample to include only exogenously determined CEO departures and their corresponding matched control observations. In addition, conducting our analysis using a subset of exogenous CEO turnovers should further alleviate endogeneity concerns arising from reverse causality or omitted variables.

Rows 2–4 of Panel B in Table 11 present the results of this analysis. In row 2, we follow Fee et al. (2013) and define exogenous turnovers as those resulting from the death or illness of outgoing CEOs or natural retirements.³¹

TABLE 11
Propensity Score Matched Sample Analysis

Table 11 reports results investigating the impact of CEO turnovers on supplier sales from a propensity-score matched sample using public suppliers and customers identified from Compustat and ExecuComp over 1993–2011. Panel A reports univariate results comparing variables across matched samples. Panel B tabulates the change in PCT_SALES due to a customer replacing its CEO in a difference-in-differences nonregression setting. PCT_SALES is sales to a customer divided by the supplier's total sales (in percent). We calculate the difference in PCT_SALES for the treatment suppliers before and after the turnover event minus the difference in PCT_SALES for the control suppliers before and after the turnover event. Statistical significance is only tabulated for the mean of the difference-in-differences. Panel C reports multivariate results from regressions with only year and customer-supplier fixed effects as control variables. Panel D reports multivariate results from regressions with the full set of control variables. The dependent variable in models 1–4 of Panels C and D is the natural logarithm of 1 plus PCT_SALES. POST_TURNOVER is an indicator variable set to 1 for the years after a customer replaces its CEO, and 0 otherwise. Row/model 1 uses the full sample of matched observations, and rows/models 2–4 restrict the sample to only customer-supplier relationships whose customer has an exogenous turnover and their corresponding matched observations. Row/model 2 includes death and health-related turnovers and turnovers where the CEO is at least 63 years old and industry-adjusted ROA is above the sample median in a given year. Row/model 3 includes death and health-related turnovers and turnovers where the CEO is at least 63 years old and industry-adjusted ROA is in the top quartile in a given year. Row/model 4 includes death and health-related stock returns are above the sample median in a given year. We industry adjust customer ROA by subtracting the median ROA of all firms in the customer's same 2-digit SIC industry. We industry adjust customer stock returns by subtracting the return on an index consisting of equal-weighted returns of all firms in the customer's same 2-digit SIC industry. Supplier control variables include REL_LENGTH, ROA, SALES_GROWTH, ln(SALES), VOLATILITY, ln(FIRM_AGE), SG&A, R&D, CAPEX, LEVERAGE, DIVIDEND_PAYER, and BLOCKHOLDER. Customer control variables include ln(CUST_SALES), CUST_ROA, CUST_R&D, CUST_LEVERAGE, CUST_ΔASSETS, and CUST_STOCK_RETURN. Table 3 provides definitions of control variables. Industries are defined at the 2-digit SIC level. Continuous variables are winsorized at their 1st and 99th percentiles each year, and dollar values are expressed in 2009 dollars. In Panel A, *, **, and *** in the column labeled Mean (Median) indicate significance at the 10%, 5%, and 1% levels, respectively, for a *t*-test for whether the two samples have equal means (a χ^2 test corrected for continuity for whether the two samples are drawn from populations with the same median). In Panels C and D, standard errors are corrected for heteroskedasticity and clustering at the supplier level (*t*-statistics are reported in parentheses). *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A. Comparison of Averages of Variables across Matched Samples

Variable	Treatment Group (no. of obs. = 4,167)		Control Group (no. of obs. = 4,167)	
	Mean	Median	Mean	Median
<i>Supplier and Customer Control Variables</i>				
REL_LENGTH	3.17**	2.00***	2.98	2.00
ROA	0.03	0.10***	0.02	0.09
SALES_GROWTH	0.26	0.10*	0.25	0.12
ln(SALES)	4.97	4.93	4.97	4.89
VOLATILITY	0.73	0.65	0.73	0.65
FIRM_AGE	2.09	2.20	2.09	2.20
SG&A	0.31	0.25	0.31	0.25
R&D	0.09	0.04	0.09	0.03
CAPEX	0.06	0.04	0.06	0.04
LEVERAGE	0.19	0.12	0.19	0.12
DIVIDEND_PAYER	0.15	0.00	0.15	0.00
BLOCKHOLDER	0.11	0.00	0.10	0.00
ln(CUST_SALES)	10.00	10.18	10.04	10.25
CUST_ROA	0.15	0.15	0.15	0.14
CUST_R&D	0.04	0.01	0.04	0.01
CUST_LEVERAGE	0.23**	0.22***	0.24	0.24
CUST_ΔASSETS	0.13**	0.07	0.11	0.07
CUST_STOCK_RETURN	0.21	−0.03***	0.18	0.02

(continued on next page)

³¹ The cleanest sample of CEO turnovers that would be free from most endogeneity concerns would be those that are the result of unexpected deaths. In our propensity score matched sample of 391

TABLE 11 (continued)
Propensity Score Matched Sample Analysis

Panel B. Difference-in-Differences Change in PCT_SALES

	Variable	Mean Treatment Diff (post-turnover – pre-turnover)	Mean Control Diff (post- turnover – pre-turnover)	Mean of Diff- in-Diffs (treatment diff – control diff)	t-Statistic for Diff-in- Diff
1	Full sample of matched observations Sample includes 694 treatment suppliers	–6.634%	–0.788%	–5.846%	–5.94***
2	Turnovers due to death, poor health, or retirement (and industry-adjusted ROA is above the sample median in a given year) Sample includes 143 treatment suppliers	–6.731%	–0.889%	–5.842%	–2.68***
3	Turnovers due to death, poor health, or retirement (and industry-adjusted ROA is in top quartile in a given year) Sample includes 83 treatment suppliers	–8.284%	0.681%	–8.965%	–3.00***
4	Turnovers due to death, poor health, or retirement (and industry-adjusted ROA and stock returns are above the sample median in a given year) Sample includes 76 treatment suppliers	–8.278%	–1.067%	–7.211%	–2.23**

Panel C. Multivariate Regression Results Excluding Customer and Supplier Control Variables

Variable	Turnovers Due to Death, Poor Health, or...			
	Full Sample of Matched Observations	Retirement and Industry-Adjusted ROA Is above the Sample Median	Retirement and Industry-Adjusted ROA Is in the Top Quartile	Retirement and Industry-Adjusted ROA and Stock Returns Are above the Sample Median
	1	2	3	4
POST_TURNOVER	–0.262*** (–3.71)	–0.330** (–2.17)	–0.373* (–1.84)	–0.562** (–2.56)
Controls	No	No	No	No
Year and customer-supplier FEs	Yes	Yes	Yes	Yes
No. of obs.	8,334	1,754	1,036	832
Adj. R ²	0.400	0.443	0.477	0.481

Panel D. Multivariate Regression Results Including Customer and Supplier Control Variables

Variable	1	2	3	4
POST_TURNOVER	–0.181*** (–2.68)	–0.379** (–2.56)	–0.422** (–2.01)	–0.568*** (–2.60)
Controls	Yes	Yes	Yes	Yes
Year and customer-supplier FEs	Yes	Yes	Yes	Yes
No. of obs.	8,334	1,754	1,036	832
Adj. R ²	0.426	0.461	0.505	0.529

Natural retirements are departures where the outgoing CEO is at least 63 years old and the firm has industry-adjusted ROA greater than the sample median in the year preceding the turnover. We industry adjust customer ROA by subtracting average industry ROA, computed as the median ROA of all firms in the customer's 2-digit SIC industry. This sample consists of 143 treatment suppliers and their respective matched control suppliers. The results in row 2 continue to show that, relative to control suppliers, there is a statistically significant

turnovers, we find only seven turnovers that result from the death of a CEO. Furthermore, only three of these seven are considered "unexpected," where we follow Johnson, Magee, Nagarajan, and Newman (1985) and Faccio and Parsley (2009) and define unexpected deaths as those that result from events such as vehicle accidents, heart attacks, strokes, suicides, assassinations, and brief illnesses.

decrease in PCT_SALES following the customer CEO turnover event for treatment suppliers.

In rows 3 and 4 of Table 11, we apply more stringent criteria for defining natural retirements. In particular, in addition to the age requirement, we require that customers have industry-adjusted ROA in the top quartile the year before the turnover in row 3, resulting in a sample of 85 treatment suppliers and their respective control suppliers. In row 4, in addition to requiring that customers have above-median industry-adjusted ROA and CEOs be at least 63 years old, we also require that customers have above-median industry-adjusted 1-year stock returns the year before the turnover. We industry adjust customer stock returns by subtracting the returns of an index consisting of equal-weighted returns of all firms in the customer's 2-digit SIC industry. This sample consists of 76 treatment suppliers. The results in rows 3 and 4 show that there is a statistically significant decrease in PCT_SALES following customer CEO turnovers for treatment suppliers relative to control suppliers.

Panels C and D of Table 11 present multivariate results from regressing $\ln(1 + \text{PCT_SALES})$ on the POST_TURNOVER indicator variable. Panel C controls for year and customer-supplier fixed effects, and Panel D includes the full set of customer and supplier control variables. Model 1 in Panels C and D presents results from using the full sample of 4,167 customer-supplier years for both the treatment and control groups. Models 2–4 in both panels present results from using the subset of exogenous CEO turnover samples that were used in rows 2–4 of Panel B. In these tests, the subsamples in models 2–4 consist of 114, 71, and 61 unique customer CEO turnovers, respectively. The results in all four models show that, after correcting for endogeneity concerns in a multivariate framework, supplier sales to customers continue to decline following CEO departures. Overall, the results in the table are consistent with customer-firm CEO turnovers causally disrupting relationships with dependent suppliers.

Last, we examine the timing of declines in supplier sales to customers around exogenous CEO turnover events and report the results of this analysis in Table 12. If sales begin to significantly decline before the turnover event, this would suggest that these turnovers are not exogenous. To examine whether there is a trend of declining sales to customers before the exit, we decompose our POST_TURNOVER variable into the following variables. $\text{POST_TURNOVER}^{-3}$, $\text{POST_TURNOVER}^{-2}$, and $\text{POST_TURNOVER}^{-1}$ are indicator variables set to 1 if a supplier's customer will replace its CEO in 3 years, 2 years, and 1 year, respectively. POST_TURNOVER^0 , POST_TURNOVER^1 , POST_TURNOVER^2 , and $\text{POST_TURNOVER}^{+3}$ are indicator variables set to 1 if a supplier's customer replaced its CEO in the current year, 1 year, 2 years, and 3 or more years ago, respectively. Thus, each variable compares the incremental difference in sales relative to the level of sales prior to 3 years before the turnover event. The results suggest that, across the three samples of exogenous turnovers, there is no significant decline in sales before the customer replaces its CEO. Yet there are large drops in sales following the turnover event.³²

³²The significant decline in sales in the first and second years after the turnover does not extend through 3 or more years after the turnover. One reason for this pattern could be due to the structure

TABLE 12
The Timing of Declines in Supplier Sales to Customers (exogenous sample)

Table 12 reports results investigating the timing of supplier sales drops relative to CEO turnovers for the propensity score matched sample (Table 11) using public suppliers and customers identified from Compustat and ExecuComp over 1993–2011. The dependent variable in models 1–3 is the natural logarithm of 1 plus PCT_SALES. PCT_SALES is sales to a customer divided by the supplier's total sales (in percent). The samples are restricted to only customer–supplier relationships whose customer has an exogenous turnover and their corresponding matched observations. Model 1 includes death and health-related turnovers and turnovers where the CEO is at least 63 years old and industry-adjusted ROA is above the sample median in a given year. Model 2 includes death and health-related turnovers and turnovers where the CEO is at least 63 years old and industry-adjusted ROA is in the top quartile in a given year. Model 3 includes death and health-related turnovers and turnovers where the CEO is at least 63 years old and industry-adjusted ROA and industry-adjusted stock returns are above the sample median in a given year. We industry adjust customer ROA by subtracting the median ROA of all firms in the customer's same 2-digit SIC industry. We industry adjust customer stock returns by subtracting the return on an index consisting of equal-weighted returns of all firms in the customer's same 2-digit SIC industry. POST_TURNOVER⁻³, POST_TURNOVER⁻², and POST_TURNOVER⁻¹ are indicator variables set to 1 if a supplier's customer will replace its CEO in 3 years, 2 years, and 1 year, respectively, and 0 otherwise. POST_TURNOVER⁰, POST_TURNOVER¹, and POST_TURNOVER² are indicator variables set to 1 if a supplier's customer replaced its CEO this year, 1 year, and 2 years ago, respectively, and 0 otherwise. POST_TURNOVER⁺³ is an indicator variable set to 1 if a supplier's customer replaced its CEO 3 or more years ago, and 0 otherwise. Supplier control variables include REL_LENGTH, ROA, SALES_GROWTH, ln(SALES), VOLATILITY, ln(FIRM_AGE), SG&A, R&D, CAPEX, LEVERAGE, DIVIDEND_PAYER, and BLOCKHOLDER. Customer control variables include ln(CUST_SALES), CUST_ROA, CUST_R&D, CUST_LEVERAGE, CUST_ASSETS, and CUST_STOCK_RETURN. Table 3 provides definitions of control variables. Continuous variables are winsorized at their 1st and 99th percentiles each year, and dollar values are expressed in 2009 dollars. Standard errors are corrected for heteroskedasticity and clustering at the supplier level (*t*-statistics are reported in parentheses). *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Variable	Turnovers Due to Death, Poor Health, or Retirement (and industry-adjusted ROA is above the sample median in a given year) 1	Turnovers Due to Death, Poor Health, or Retirement (and industry-adjusted ROA is in the top quartile in a given year) 2	Turnovers Due to Death, Poor Health, or Retirement (and industry-adjusted ROA and stock returns are above the sample median in a given year) 3
POST_TURNOVER ⁻³	-0.078 (-0.40)	0.492 (1.59)	-0.037 (-0.10)
POST_TURNOVER ⁻²	-0.178 (-0.76)	-0.165 (-0.50)	-0.361 (-1.14)
POST_TURNOVER ⁻¹	-0.337 (-1.28)	-0.480 (-1.25)	-0.495 (-1.18)
POST_TURNOVER ⁰	-0.423 (-1.42)	-0.527 (-1.24)	-0.528 (-1.03)
POST_TURNOVER ¹	-0.831** (-2.38)	-1.063** (-2.15)	-1.086* (-1.93)
POST_TURNOVER ²	-0.814** (-2.01)	-0.996 (-1.63)	-1.358** (-2.03)
POST_TURNOVER ⁺³	-0.759 (-1.57)	-0.839 (-1.26)	-1.335* (-1.68)
Controls	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes
Customer–supplier FEs	Yes	Yes	Yes
No. of obs.	1,754	1,036	832
Adj. R ²	0.464	0.520	0.533

B. Other Robustness Tests of the Main Specification

We next address the concern that POST_TURNOVER simply captures the pattern that all relationships eventually decline and come to an end. While we control for the length of relationships in our models, this control variable may not completely account for this concern because control and treatment group

of our panel data. Because customer–supplier relationships exit our sample beginning 2 years after the relationship ends, the only observations remaining 3 or more years after the turnover are those in which the relationship does not end, creating a mechanical increase in sales in the later years after the turnover. However, this mechanical increase has a smaller effect on our main results because there are fewer observations in these later years.

relationships are not perfectly matched in terms of the timing of relationship initiations. Thus, if POST_TURNOVER simply picks up this pattern, we would expect to find a negative relation between POST_TURNOVER and sales to customers, regardless of how this variable is defined with respect to the timing of the turnover. To test whether this explanation holds, we create two placebo tests in which POST_TURNOVER is redefined to take a value of 1 beginning 2 years after (2 years before) the actual turnover event, with results shown in model 1 (2) of Table 13. We find a positive relation between POST_TURNOVER and PCT_SALES when the post turnover window is not properly defined. Thus, the results from these two tests help alleviate the concern that our POST_TURNOVER variable simply proxies for the natural decline in relationships.

As an alternative way to address this concern, we create three different matched samples that explicitly match on relationship length. We use these samples to rerun our main model specification and report the results in models 3–5 of Table 13. We note that the number of observations used in these tests is significantly lower as a result of our matching procedure, described below. In model 3, we match, with replacement, treatment and control observations on relationship length, fiscal year, customer industry, and supplier industry. For each customer–supplier observation in the treatment group with multiple matches, we select the closest matched observation in terms of the smallest distance based on the norm function $\| \cdot \|$ and using the natural logarithm of customer and supplier total sales

TABLE 13
Other Robustness Tests of the Main Specification

Table 13 reports results from OLS fixed effects regressions using public suppliers and customers identified from Compustat and ExecuComp over 1993–2011. The dependent variable in models 1–5 is the natural logarithm of 1 plus PCT_SALES. PCT_SALES is sales to a customer divided by the supplier's total sales (in percent). Models 1 and 2 use the original sample created by matching each ExecuComp customer to its suppliers from the first year of the customer–supplier relationship to 2 years after the relationship ends. In model 1, POST_TURNOVER is an indicator variable set to 1 beginning 2 years after a customer replaces its CEO, and 0 otherwise. In model 2, POST_TURNOVER is an indicator variable set to 1 beginning 2 years before a customer replaces its CEO, and 0 otherwise. Models 3–5 use samples of matched treatment and control observations using three different matching procedures that focus on matching on relationship length. Model 3 matches on relationship length, fiscal year, and both customer and supplier industries. Model 4 uses the same matching procedure as model 3 and also matches on customer and supplier total sales and customer and supplier ROA. Model 5 matches each customer–supplier relationship in the treatment group to the same supplier but different customer in the control group in the same year with the same relationship length. Supplier control variables include REL_LENGTH, ROA, SALES_GROWTH, ln(SALES), VOLATILITY, ln(FIRM_AGE), SG&A, R&D, CAPEX, LEVERAGE, DIVIDEND_PAYER, and BLOCKHOLDER. Customer control variables include ln(CUST_SALES), CUST_ROA, CUST_R&D, CUST_LEVERAGE, CUST_ASSETS, and CUST_STOCK_RETURN. Table 3 provides definitions of control variables. Continuous variables are winsorized at their 1st and 99th percentiles each year, and dollar values are expressed in 2009 dollars. Standard errors are corrected for heteroskedasticity and clustering at the supplier level (*t*-statistics are reported in parentheses). *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Variable	Setting POST_TURNOVER to 1 Starting 2 Years after the Turnover 1	Setting POST_TURNOVER to 1 Starting 2 Years before the Turnover 2	Match on Relationship Length (match 1) 3	Match on Relationship Length (match 2) 4	Match on Relationship Length (match 3) 5
POST_TURNOVER	0.095** (2.35)	0.360*** (7.37)	−0.260*** (−4.49)	−0.254*** (−3.11)	−0.195** (−2.32)
Controls	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes
Customer–supplier FEs	Yes	Yes	Yes	Yes	Yes
No. of obs.	22,797	22,797	10,228	4,060	2,682
Adj. <i>R</i> ²	0.414	0.416	0.547	0.578	0.490

and customer and supplier ROA.³³ Model 4 uses the same matching procedure, but we first require matched pairs to be within 1 standard deviation of the natural logarithm of customer and supplier total sales and customer and supplier ROA before minimizing the distance when there are multiple matches.

Last, in model 5 of Table 13, we match each customer–supplier relationship in the treatment group to the same supplier but with a different customer in the control group and require that each matched pair have the same fiscal year and the same relationship length. This method forces matched pairs to have the exact same supplier characteristics because the supplier is the same for each treatment and control observation. Similarly, for each observation with multiple matches, we select the closest matched observation with the smallest distance based on the natural logarithm of customer total sales and customer ROA.³⁴ Overall, in all matched samples, we continue to find a negative relation between customer CEO departures and supplier sales. Thus, these additional findings provide further evidence that our results do not simply capture the pattern that all relationships eventually decline and end.

VI. Conclusion

Suppliers that depend on one or more customers for a substantial portion of their sales are less diversified and often make RSIs that offer little value outside of these relationships. As a result, disruptions in relationships with these customers represent a significant source of risk for these suppliers and their investors. We provide evidence that the replacement of a customer's CEO significantly disrupts its relationships with dependent suppliers. Our findings show that dependent suppliers experience a significant decline in sales to customers that replace their CEOs. We also document that supplier firm value and operating performance deteriorate following CEO turnovers and that supplier shareholders react negatively to the announcement of a customer CEO departure. In contrast, when major customers face greater barriers to switching suppliers and when suppliers make fewer RSIs, suppliers do not experience a loss in sales to customers that replace their CEOs.

³³The norm distance function method is discussed by Roberts and Whited (2013). All matched control variables are first standardized to have a mean of 0 and a standard deviation of 1. We then minimize the following distance value for each matched pair with multiple matches:

$$d(X, Z) = \sqrt{\sum_{i=1}^4 (x_i - z_i)^2},$$

where x_i indicates, for the treatment group i) the natural logarithm of supplier total sales, ii) the natural logarithm of customer total sales, iii) supplier ROA, and iv) customer ROA. Similarly, z_i indicates the same set of variables for the control group.

³⁴This method of matching also provides further evidence that our results are not driven by potential biases related to the 10% reporting requirement. Specifically, we expect some consistency in a supplier's disclosures across customers. Thus a supplier that discloses sales to one customer just under the 10% threshold is likely to also disclose sales to other customers that are just under the threshold in the same year. Consequently, treatment and control groups should have similar biases, and our difference-in-differences methodology should difference away these biases.

We also find larger disruptions in customer–supplier relationships when incumbents are replaced with outsiders, are forced out of office, or have more discretionary decision-making power. Successors are also more likely to divest assets and business segments in their first few years in office. These findings are consistent with the idea that much of the decline in supplier sales results from successors abandoning the strategies of entrenched incumbents.

Our findings may have important implications for suppliers and their risk management practices. For example, suppliers should consider the likelihood of near-term management turnover before entering into long-term relationships with customers. They should be especially concerned when customers can more easily switch suppliers and when suppliers must make significant RSIs that will provide little value if the relationship ends. In addition, before entering a relationship, suppliers should consider the tenure and current age of the customer's CEO and determine whether the prospective customer has a succession plan in place. A lack of a clear frontrunner or set of internal candidates identified as competing for the top post may indicate that the customer's intent is to hire an outsider, which would signal a higher probability of the relationship ending once the outsider is hired. An evaluation of the customer's succession process, therefore, can impact the supplier's ex ante valuation of these relationships. This notion is consistent with a PricewaterhouseCoopers article noting that the lack of a thorough succession plan can create uneasiness among a firm's suppliers, which can harm the firm's financial health.³⁵

While we recognize that our inferences are limited to the effects of customer CEO turnovers on dependent suppliers, over 46% of all public U.S. industrial companies relied on at least one customer for at least 10% of total revenues in 2011. To the extent that the effects of CEO turnovers differ for nondependent suppliers or new suppliers, the generalizability of our findings to all suppliers may be limited. An examination of the effects of CEO turnovers on nondependent suppliers is beyond the scope of our paper, and we leave this analysis for future research.

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³⁵See "Succession Planning: Delaying Could Cost You," *Growing Your Business*, Vol. 64. PricewaterhouseCoopers, London, 2012.

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